Role of Pyridinic Nitrogen in Redox Activity of Nitrogen-Doped Carbon Fabrics

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Highlights

- Nitrogen configurations tuned via g-C_3N_4-assisted carbonization
- Pyridinic-N identified as a major electrochemically active nitrogen configuration
- Optimized carbon fabric shows high capacitance and cycling stability
- DFT reveals favorable proton adsorption at pyridinic-N sites
- Experimental and theoretical insights into structure–electrochemical property relationships

ABSTRACT

Controlling the configuration of nitrogen dopants is crucial for optimizing the electrochemical behavior of carbon materials. Here, we present a solid-state strategy that employs exfoliated g-C_3N_4 nanosheets as both a nitrogen source and porogen to synthesize nitrogen-doped carbon fabrics (NCF-T, T = 600–1000 °C). By adjusting the pyrolysis temperature, the relative proportions of pyridinic, pyrrolic, and graphitic nitrogen species were systematically tuned while preserving the textile-like morphology. Among the series, NCF-900 exhibited an optimal balance of electrochemically accessible pyridinic-N sites, abundant defect sites, and a well-developed mesoporous network, resulting in high specific capacitance (139.6 F g^{-1} at 0.5 A g^{-1}), excellent rate capability, and 89 % capacitance retention after 10,000 cycles. Complementary density functional theory (DFT) calculations revealed that pyridinic nitrogen sites possess the most favorable proton adsorption energy and exhibit pronounced charge redistribution near the Fermi level, consistent with their superior redox activity. These combined experimental and theoretical results identify pyridinic-N as a major contributor to redox activity, facilitating reversible proton-coupled electron transfer and enhanced ion adsorption. Overall, this work provides experimental and theoretical insight into the relationship between nitrogen configuration and charge-storage behavior, while demonstrating a scalable, gas-free route for fabricating flexible, high-performance carbon electrodes for next-generation supercapacitors.

1. Introduction

Carbon-based materials are widely recognized as indispensable electrode candidates for next-generation electrochemical energy-storage devices. This is due to their excellent electrical conductivity, large surface area, strong chemical stability, and robust mechanical strength [1], [2], [3]. Their electrochemical performance can be further enhanced by heteroatom doping, without compromising structural integrity. Among the various dopants, nitrogen is especially attractive. Its atomic size and high electronegativity are similar to carbon, which allows it to integrate easily into the carbon lattice and alter the local charge distribution [4], [5], [6].

The specific configuration of nitrogen atoms—typically categorized as pyridinic, pyrrolic, graphitic, and oxidized nitrogen—plays a decisive role in determining the electronic properties and electrochemical behavior of carbon materials. The representative nitrogen configurations and their bonding characteristics are illustrated in Figure 1 and summarized in Table 1. For instance, graphitic nitrogen improves electronic conductivity by acting as a substitutional dopant. Pyrrolic nitrogen adds redox activity and helps generate porosity. Pyridinic nitrogen is typically located at edges and defect sites. It provides lone pair electrons, increases local electron density, and promotes electrolyte-ion adsorption as well as redox reactions. Consequently, pyridinic-N is often regarded as one of the most electrochemically active nitrogen configurations in carbon materials [7], [8], [9]. Despite these benefits, systematic studies that quantitatively link each nitrogen configuration to electrochemical performance appear to be still lacking. This is mainly due to challenges in controlling nitrogen configurations during synthesis. Such a knowledge gap limits the rational design of high-performance carbon electrodes.

Traditionally, ammonia- or urea-assisted methods have been widely used for nitrogen doping of carbon materials [10], [11], [12], [13], [14]. However, these methods often result in uncontrolled gas release and the random formation of pores or defects. This leads to imprecise nitrogen incorporation and poor reproducibility. In contrast, graphitic carbon nitride (g-C_3N_4) is a promising alternative as a nitrogen source. Solid-state doping with g-C_3N_4 allows more precise control of the crystalline framework and morphology. As a result, nitrogen can be introduced in a more controllable manner while preserving the structural integrity of the carbon framework. Nitrogen introduced via this route typically enables better control over the formation of different nitrogen configurations, including pyridinic and graphitic configurations [15]. Moreover, $g\text{-}C_3N_4$-based doping avoids the safety and environmental problems caused by toxic gas release, making it a more scalable and environmentally friendly process.

In this study, we employed exfoliated $g\text{-}C_3N_4$ nanosheets as both the nitrogen source and sacrificial porogen to synthesize nitrogen-doped carbon fabrics (NCF-T, T = 600–1000 °C) through a simple solid-state route. By varying the pyrolysis temperature, the distributions of pyridinic, pyrrolic, and graphitic nitrogen species were systematically tuned while maintaining the textile-like morphology of the carbon framework. The sample prepared at 900 °C (NCF-900) exhibited the most favorable combination of a high density of electrochemically accessible pyridinic-N sites, defect-rich structure, and mesoporosity, resulting in superior capacitive performance and durability. Furthermore, density functional theory (DFT) calculations were conducted to elucidate the site-dependent electrochemical activity of different nitrogen configurations and to clarify the role of pyridinic-N in proton-coupled charge-storage processes. This work provides both experimental and theoretical insight into how specific nitrogen configurations govern charge storage, offering design guidance for scalable, high-performance carbon-based supercapacitor electrodes.

2. Experimental

2.1. Fabrication of bulk $g\text{-}C_3N_4$

Bulk graphitic carbon nitride ($g\text{-}C_3N_4$) was synthesized by the direct thermal polymerization of melamine in air. Typically, 10 g of melamine powder (≥99%, Sigma-Aldrich) was placed in a covered alumina crucible and heated in a muffle furnace at 550 °C for 4 h with a ramping rate of 5 °C min^{-1}. After cooling naturally to room temperature, the obtained light-yellow solid was ground into a fine powder using an agate mortar. The resulting material, denoted as bulk $g\text{-}C_3N_4$, was stored in a desiccator prior to use.

2.2. Fabrication of g-C_3N_4-coated cotton fabric (CF/g-C_3N_4)

To prepare the g-C_3N_4-coated cotton precursor, 100 mg of bulk g-C_3N_4 powder was dispersed in 30 mL of ethanol using ultrasonication (40 kHz, 100 W) for 2 h to form a homogeneous yellow suspension. A rectangular piece of cotton fabric (approximately 2 × 4 cm^2, denoted as CF) was then immersed in the dispersion and further sonicated for 1 h to ensure thorough infiltration and coating. After sonication, the fabric was removed, gently squeezed to remove excess liquid, and dried in a convection oven at 80 °C for 12 h until a constant mass was achieved. The resulting CF/g-C_3N_4 composite exhibited a uniform surface coating of exfoliated g-C_3N_4 nanosheets, with an estimated loading of ~5 mg cm^{-2} based on mass difference measurements.

2.3. Preparation of nitrogen-doped carbon fabrics (NCF-T)

The Cotton Fabric/g-C_3N_4 precursor was converted into nitrogen-doped carbon fabrics (NCF-T, where T = 600–1000 °C) through pyrolysis under a controlled atmosphere. Each sample was placed in an alumina crucible and transferred into a quartz-tube furnace (Lindberg/Blue M STF55433C) evacuated to ~0.1 Torr, followed by purging with high-purity argon (99.999%) at a flow rate of 150 sccm. The furnace temperature was ramped at 10 °C min^{-1} to the target temperature (600, 700, 800, 900, or 1000 °C) and maintained for 1 h. After annealing, the system was naturally cooled to room temperature under continuous Ar flow. The obtained black, flexible fabrics were designated as NCF-600, NCF-700, NCF-800, NCF-900, and NCF-1000, respectively.

2.4. Physical and chemical characterization

The morphology and microstructure of the samples were characterized by field-emission scanning electron microscopy (FE-SEM, JEOL JSM-7000F) equipped with an energy-dispersive X-ray spectrometer (EDX, Oxford AZtec X-Max), as well as transmission electron microscopy (TEM, JEOL JEM-2100 operated at 200 kV) combined with selected area electron diffraction (SAED). Raman spectroscopy was performed using a Renishaw inVia Raman microscope with a 514 nm excitation laser, 1 μm spot size, and a ×50 objective lens to evaluate the structural disorder and graphitic characteristics of the carbon framework. Crystal structures were analyzed by X-ray diffraction (XRD, Panalytical Empyrean, Cu Kα radiation, λ = 1.5406 Å). Nitrogen adsorption–desorption isotherms were measured at 77 K using a Micromeritics TriStar II 3020 analyzer to determine the specific surface area (BET method) and pore-size distribution (BJH method), after degassing the samples at 120 °C for 12 h under vacuum. Surface chemical composition and bonding states were investigated by X-ray photoelectron spectroscopy (XPS, MultiLab 2000, Thermo Fisher Scientific, Al Kα source). The binding energies were calibrated using the C 1s peak at 284.8 eV, and the spectra were analyzed by Gaussian–Lorentzian peak fitting.

2.5. Electrochemical measurements

The electrochemical properties of the nitrogen-doped carbon fabrics were evaluated using a Bio-Logic VSP workstation in both three-electrode and symmetric two-electrode configurations. For the three-electrode measurements, the NCF-T samples were directly used as binder-free working electrodes, with a platinum coil and an Ag/AgCl electrode as the counter and reference electrodes, respectively, in a 1 M H_2SO_4 aqueous electrolyte. Cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS) were conducted within a potential window of 0–1.0 V (vs. Ag/AgCl), with EIS measured from 100 kHz to 50 mHz using a 10 mV AC perturbation. For the symmetric two-electrode configuration, two identical pieces of NCF-T (mass loading ~1 mg cm^{-2}) were assembled as binder-free electrodes using 6 M KOH as the electrolyte and an ashless filter paper separator in a Hohsen HS-Flat Cell. CV was recorded at scan rates of 5–1000 mV s^{-1}, GCD tests were performed at current densities of 0.3–10 A g^{-1}, and EIS was measured under identical conditions. Cycling stability of NCF-900 was evaluated by continuous CV cycling over 5,000 cycles (three-electrode system) and 10,000 cycles (two-electrode system). The specific capacitance of working electrode in the three-electrode system ($C_{sp,3E}$) and specific capacitance of symmetric two electrode system ($C_{sp,2E}$) were calculated using the following equations.

$$C_{sp,3E} = \frac{I\Delta t}{m\Delta V}$$

where I (A) is the discharge current, Δt (s) is the discharge time, m (g) is the total mass of active material in working electrode, ΔV (V) is the potential window.

$$C_{sp,2E} = \frac{4I\Delta t}{m\Delta V}$$

where I (A) is the discharge current, Δt (s) is the discharge time, m (g) is the total mass of active materials in both electrodes, ΔV (V) is the operating voltage window of the device.

2.6. Fabrication and demonstration of symmetric supercapacitor device

The symmetric supercapacitor device was assembled using two circular NCF-900 electrodes (14 mm diameter) directly punched from the carbon fabric and used without any conductive additives or polymer binders. A separator soaked in 6 M KOH electrolyte was placed between the two electrodes, and the assembly was packaged in a coin-cell configuration.

2.7. Density functional theory (DFT) calculations

DFT calculations were performed using the Vienna *ab initio* Simulation Package (VASP) [16] to compare the redox activity of different nitrogen configurations (pyridinic, pyrrolic, graphitic, and oxidized nitrogen). The Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) [17] was adopted for the exchange–correlation functional. A plane-wave cut-off energy of 400 eV and Γ-point-only Brillouin zone sampling were employed for all calculations, ensuring free energy and pressure convergence within 10 meV/atom and 10 kBar, respectively. A 15 Å vacuum layer was introduced along the out-of-plane direction to avoid interactions across periodic boundaries. Structural relaxation was performed until the residual forces on each atom were below 0.01 eV/Å.

The hydrogen adsorption energy (E_{ads}) was calculated by the following equation:

$$E_{ads} = E_{H@NCF} - E_{NCF} - \frac{1}{2}E_{H_2}$$

where $E_{H@NCF}$ and E_{NCF} are the total energies of the nitrogen-doped carbon structure with and without an adsorbed hydrogen atom, respectively, and E_{H_2} is the total energy of an isolated hydrogen molecule. Bader charge analysis [18] was employed to quantify the charge transfer induced by hydrogen adsorption.

3. Results and discussion

3.1. Microstructural and chemical evolution by g-C_3N_4-mediated nitrogen doping

The synthesis route and structural evolution of nitrogen-doped carbon fabrics (NCF-T, T = 600–1000 °C) are illustrated in Figure 2a. To introduce nitrogen dopants while preserving the fibrous architecture of the cotton fabric, a solid-state approach was employed using exfoliated g-C_3N_4 nanosheets as both the nitrogen source and sacrificial template. The physical appearance of the samples is shown in Figure S1a, while EDS mapping confirms the successful incorporation of nitrogen species within the carbon fabric (Figure S1b). To further examine the spatial distribution of nitrogen species within the carbon fibers, cross-sectional EDS analysis was performed on NCF-900 (Figure S1c). A distinct nitrogen signal was detected near the outer region of the fiber, while the nitrogen intensity decreased markedly toward the interior and became negligible in the fiber core. This result indicates that the nitrogen functionalities introduced by the g-C_3N_4-assisted treatment are predominantly concentrated in the surface and near-surface regions of the carbon fibers. Thermogravimetric analysis (Figure S2) revealed two distinct mass-loss events corresponding to cellulose decomposition at approximately 350 °C and g-C_3N_4 degradation from 500 °C to 700 °C [19], [20]. The evolution of nitrogen configurations in NCF-T with increasing pyrolysis temperature reflects the progressive transformation of g-C_3N_4-derived nitrogen into lattice-incorporated species within the carbon framework. Distinct XPS signals corresponding to pyridinic, pyrrolic, graphitic, and oxidized nitrogen appeared (Figure 2b–d), confirming the conversion of molecular g-C_3N_4 into lattice-incorporated nitrogen species [21], [22]. The relative percentages of nitrogen configurations in the NCF-T series are summarized in Table 2, while the corresponding total nitrogen contents and calculated atomic contents of each nitrogen configuration are presented in Table 3. As the pyrolysis temperature increases from 600 to 1000 °C, the total nitrogen content gradually decreases, accompanied by temperature-dependent evolution of nitrogen configurations, reflecting thermally induced transformation among nitrogen species. These transformations may lead to higher structural disorder and the formation of electrochemically active sites [23]. Notably, the calculated pyridinic-N content alone does not fully explain the electrochemical performance, indicating that the accessibility and chemical environment of nitrogen species are also important factors governing charge storage. At the same time, a gradual rise in graphitic nitrogen could contribute to enhanced electronic conductivity [21]. Notably, the oxidized-N component in NCF-600 appears at a binding energy comparable to those observed in NCF-800, NCF-900, and NCF-1000 (~403 eV), suggesting a similar chemical origin associated with surface oxidation or weakly oxygen-coordinated nitrogen species. In contrast, NCF-700 exhibits a pronounced shift of the oxidized-N peak toward higher binding energy (~404 eV), indicating the presence of more strongly oxidized or electronically perturbed nitrogen species. This behavior is consistent with the transitional nature of the carbon framework at 700 °C, where g-C_3N_4 has largely decomposed but the carbon lattice has not yet fully reorganized, leading to metastable nitrogen environments and enhanced electron-withdrawing effects.

The deconvoluted N 1s spectrum of g-C_3N_4 is characterized by three distinct components located at ~398.1 eV (C=N–C), ~399.4 eV ($(C)_3$–N), and ~400.7 eV (N–H) [24], which are well-recognized fingerprints of the polymeric g-C_3N_4 framework (Figure S3). The presence of these multiple g-C_3N_4-related nitrogen configurations indicates incomplete thermal decomposition at 600 °C, consistent with the TGA results [25]. Under these conditions, a substantial fraction of the nitrogen species remains associated with the polymeric g-C_3N_4 framework rather than being incorporated into a conductive carbon lattice. Consequently, part of the ~398 eV signal in NCF-600 is likely associated with residual g-C_3N_4-derived C=N moieties. Therefore, the high pyridinic-N fraction observed in NCF-600 is attributed, at least in part, to residual g-C_3N_4-derived species rather than exclusively to lattice-incorporated pyridinic-N. Although NCF-700 exhibits a relatively high fraction of pyridinic-N according to XPS analysis, TGA results indicate that 700 °C represents a transitional temperature at which the decomposition of g-C_3N_4 is nearly completed but the nitrogen species have not yet fully reconstructed into electrochemically effective configurations. As a result, a significant portion of pyridinic-N in NCF-700 is likely associated with residual or partially reorganized g-C_3N_4-derived C=N moieties or poorly connected carbon domains, which are less accessible for proton-coupled redox reactions. It should be noted that the N 1s binding energy of pyridinic-N (~398.3–398.8 eV) overlaps significantly with the C=N–C moieties of g-C_3N_4 (~398.1 eV), making it difficult to unambiguously distinguish lattice-incorporated pyridinic-N from residual g-C_3N_4-derived species by XPS alone. Therefore, the elevated pyridinic-N fraction observed in low-temperature samples does not directly reflect the density of structurally integrated pyridinic sites within the carbon framework. Instead, only pyridinic-N species effectively incorporated into the carbon lattice are expected to contribute significantly to the functional properties of the material. For samples pyrolyzed at 800–1000 °C, pyridinic nitrogen exhibits a clear temperature-dependent evolution. At 800 °C, a substantial fraction of pyridinic-N is already established, indicating effective conversion of nitrogen from molecular precursors into edge-located configurations within the carbon lattice. Upon increasing the temperature to 900 °C, the contribution of pyridinic-N reaches its maximum, suggesting that this temperature provides an optimal balance between nitrogen stabilization and defect preservation. Such edge-associated pyridinic sites are known to introduce localized electronic states and facilitate reversible proton-coupled electron transfer. When the temperature is further increased to 1000 °C, the pyridinic-N content decreases markedly, likely due to thermally driven reconstruction of edge sites and transformation of less stable nitrogen species into more energetically favorable graphitic configurations or their partial elimination from the carbon framework [26]. The high-resolution C 1s XPS spectra of pristine cotton fabric and NCF-T samples (Figure S4) further confirm the chemical evolution of carbon bonding during nitrogen doping. Each spectrum can be deconvoluted into three main peaks at approximately 284.6 eV (C–C/C=C), 286.1 eV (C–O/C–N), and 288.5 eV (C=O/C–O–N). Compared with pristine fabric, the relative intensity of C–O/C–N and C=O/C–O–N components markedly increases after g-C_3N_4-assisted carbonization, indicating successful incorporation of nitrogen into the carbon framework. The progressive enhancement of C–O/C–N bonding up to 900 °C reflects the formation of nitrogen-linked carbon species, whereas its reduction at 1000 °C suggests further graphitization and structural ordering.

The structural evolution of the nitrogen-doped carbon fabrics was examined by X-ray diffraction (XRD) and Raman spectroscopy. In the XRD patterns (Figure 3a), broad reflections at 25° and 43° correspond to the (002) and (100) planes of turbostratic carbon, characteristic of an amorphous framework [27]. As the pyrolysis temperature increased, the (002) peak gradually sharpened and shifted, suggesting improved graphitic ordering [27], [28]. The cotton fabric/g-C_3N_4 precursor exhibited an additional peak near 27.4°, confirming the presence of g-C_3N_4 [29]. In contrast, the diffraction peaks of NCF-600 to NCF-1000 become progressively weaker, indicating the decomposition of the molecular precursor and the formation of nitrogen-doped carbon domains with reduced long-range order. Raman spectra (Figure 3b) display the typical D and G bands, with NCF-900 showing the highest I_D/I_G ratio (0.87), implying a defect-rich structure induced by pyridinic and pyrrolic nitrogen incorporation (Table S1) [30]. The slight narrowing of both bands at 900 °C suggested increased ordering within the disordered carbon network [31].

Morphological analysis using scanning electron microscopy (SEM) and transmission electron microscopy (TEM) further supports this structural transition (Figure 3c). The NCFs exhibit a wrinkled, porous framework consisting of interconnected nanosheets. High-resolution TEM revealed partially graphitized regions, while the diffuse rings in the selected area electron diffraction (SAED) pattern were consistent with a turbostratic structure [32]. Elemental mapping confirms the presence of C, N, and O throughout the matrix. Among all samples, NCF-900 achieved the most balanced structure, combining defect-induced disorder with sufficient graphitic order—an architecture favorable for both electrical conductivity and ion accessibility [23]. The morphological evolution of the NCF-T samples was further examined by SEM, as shown in Figure S5. The samples carbonized at 600–1000 °C exhibit distinct temperature-dependent surface features. NCF-900 retains a well-defined fibrous morphology with a rough and porous surface, while NCF-600 appears under-carbonized with loosely bound fibers. In contrast, NCF-1000 shows partial pore collapse and smoother surfaces, indicating excessive graphitization at high temperature. These visual observations support the structural trends discussed above, confirming that 900 °C yields the optimal balance between carbonization and porosity development.

The nitrogen adsorption–desorption results (Figure 3d) provided additional insight into the porosity development driven by g-C_3N_4. All NCF samples exhibited type IV isotherms typical of mesoporous materials, but NCF-900 showed a distinct hysteresis loop and the highest specific surface area of 646.2 $m^2\ g^{-1}$ (Figure 3e), nearly twice that of the undoped control. The corresponding Barrett-Joyner-Halenda (BJH, Figure 3f) pore-size distribution revealed mesopores centered at approximately 3.9 nm, an ideal dimension for efficient ion diffusion and charge buffering [33]. By contrast, under-carbonized NCF-600 showed limited porosity, whereas over-graphitized NCF-1000 suffered from partial pore collapse.

Collectively, these observations suggested that g-C_3N_4-assisted pyrolysis enables precise control of nitrogen incorporation and mesopore formation, producing a carbon framework that unites high defect density with enhanced electrical conductivity. These structural and chemical features collectively determine the accessibility and activity of electrochemical sites, which are evaluated in the following section.

3.2. Electrochemical implications of nitrogen configurations in carbon electrodes

The capacitive behavior of the NCF-T materials was examined using both a three-electrode configuration in 1 M H_2SO_4 electrolyte and a symmetric two-electrode configuration in 6 M KOH electrolyte. In the symmetric two-electrode configuration using 6 M KOH electrolyte, the cyclic voltammetry (CV) curves maintained nearly rectangular profiles at 5 mV s^{-1}, indicative of efficient ion transport and reversible adsorption processes (Figure 4a). In contrast, the CV curves obtained in 1 M H_2SO_4 exhibited broader redox features (Figure 4b), suggesting an additional pseudocapacitive contribution associated with nitrogen-containing functional groups. Among the series, NCF-900 consistently exhibited the highest capacitance in both systems, suggesting that its favorable nitrogen configuration and porous structure are beneficial for charge storage. Corresponding charge–discharge profiles (Figure S6a) revealed longer discharge times for NCF-900 under identical current densities, whereas NCF-600, NCF-700 and NCF-1000 showed lower capacitance, likely due to incomplete nitrogen incorporation into electrochemically active lattice configurations (NCF-600 and NCF-700) or excessive graphitization (NCF-1000). Electrochemical impedance spectroscopy (EIS, Figure S6c) further supported these observations: NCF-900 displayed a smaller semicircle in the high-frequency region, implying lower charge-transfer resistance, and a steeper slope in the low-frequency region, characteristic of efficient ion diffusion [34]. The superior performance of NCF-900 is attributed to the synergistic combination of favorable nitrogen configurations, defect-rich carbon domains, and well-developed mesoporosity, which collectively enhance ion accessibility and charge-storage capability. After 10,000 charge–discharge cycles in 6 M KOH, NCF-900 retained 89.2 % of its initial capacitance (Figure S6d), suggesting good structural integrity and electrochemical stability. This stability may be related to the mechanically integrated carbon framework, strong anchoring of nitrogen dopants, and the preservation of electrochemically active sites during repeated cycling [35].

As illustrated in Figure 4c, the development of mesopores appears to be closely associated with the type and spatial distribution of nitrogen functionalities. Pyridinic and pyrrolic nitrogen species, which are often located at edge or defect sites, likely induce local lattice distortion during carbonization [36], [37]. These defective regions can act as preferential etching points that expand into mesopores upon gas evolution or activation. In contrast, graphitic nitrogen mainly enhances electronic conductivity and contributes to structural stability but has a limited role in pore formation [38]. Hence, a higher proportion of edge-type nitrogen species may facilitate the formation of mesoporous structures and promote faster ion transport, consistent with the porosity trends discussed in Figure 3.

In the three-electrode configuration using 1 M H_2SO_4 electrolyte, the CV curves exhibited broad redox humps superimposed on the capacitive background (Figure 4c), indicating the contribution of pseudocapacitive charge-storage processes associated with nitrogen-containing functional groups [39], [40]. The charge storage process of NCF-900 is therefore presumed to involve both ion adsorption and surface redox reactions. As shown schematically in Figure 4d, the graphitic domains likely provide continuous pathways for electron transport and mechanical support, while nitrogen dopants perturb the local electronic structure and create active sites for reversible proton-coupled electron transfer. Pyridinic nitrogen, located at edge sites, can donate lone-pair electrons [41], enabling the following reversible reaction:

$$–C{=}N–C– + H^{+} + e^{-} \rightleftharpoons –C{=}N(H)–C– \text{(in acidic media)}$$

Pyrrolic nitrogen, while less conductive, is generally considered to play a secondary role in pseudocapacitance, mainly through defect-associated surface states rather than direct proton-coupled redox activity [42]. In contrast, graphitic nitrogen incorporated into the basal plane enhances the intrinsic electrical conductivity by modulating the band structure and increasing the density of states near the Fermi level [43].

The porous architecture of NCF-900 complements these chemical features by providing interconnected pathways for ion transport and an abundant interfacial area for charge accumulation. Mesopores function as ion reservoirs and diffusion channels, while micropores contribute to double-layer formation [44]. The coexistence of configuration-specific nitrogen dopants and a hierarchical pore network, therefore, appears to promote fast charge transfer, high capacitance, and sustained cycling stability [39]. Additional electrochemical results in Figure S7 presents the electrochemical performance of the NCF-T electrodes measured in a three-electrode configuration using 1 M H_2SO_4 electrolyte. The GCD curves (Figure S7a) show that NCF-900 exhibits the longest discharge time, indicating the highest capacitance among the samples. The areal capacitance results (Figure S7b) further confirm this trend, with NCF-900 consistently outperforming NCF-600, NCF-700, NCF-800, NCF-1000, and the pristine cotton fabric across all current densities. At a current density of 1 mA cm^{-2}, NCF-900 exhibits a high areal capacitance of 789.4 mF cm^{-2}, demonstrating efficient charge storage capability. The inferior performance of the remaining samples can be attributed to distinct temperature-dependent limitations. Although NCF-600 exhibits a higher pyridinic-N fraction by XPS, TGA and XPS analyses suggest that a significant portion of these species originate from incompletely decomposed g-C_3N_4 rather than from electrochemically active lattice-incorporated configurations. Consequently, the apparent high pyridinic-N fraction does not directly translate into enhanced charge-storage performance. This observation further indicates that the electrochemical behavior does not simply follow the trend of total nitrogen content or the relative abundance of a specific nitrogen species alone. Instead, the electrochemical activity is governed by the combined effects of nitrogen configuration, nitrogen accessibility, and the structural characteristics of the carbon framework. For NCF-1000, the markedly reduced total nitrogen content originates from extensive nitrogen loss at elevated pyrolysis temperature. Although pyridinic-N species are still present, their absolute density becomes limited, which restricts the overall contribution of proton-coupled redox reactions. In addition, excessive graphitization at 1000 °C suppresses defect-rich edge sites and reduces ion-accessible active regions. The partial loss of mesoporosity further restricts electrolyte access to electrochemically active sites, resulting in lower pseudocapacitive charge-storage capability despite improved graphitic ordering. As a result, the capacitive performance of NCF-1000 reflects a trade-off between improved electronic transport and diminished density of electrochemically active pyridinic-N sites. The Nyquist plots (Figure S7c) reveal that NCF-900 possesses the smallest semicircle in the high-frequency region, corresponding to the lowest charge-transfer resistance and more efficient ion/electron transport at the electrode–electrolyte interface. Cycling stability tests (Figure S7d) demonstrate that NCF-900 retains 69.6 % of its initial capacitance after 5,000 cycles at a scan rate of 100 mV s^{-1}, with only slight distortion of the CV profile, suggesting satisfactory long-term durability. Overall, these results highlight that NCF-900 achieves the most balanced combination of capacitance, conductivity, and cycling stability among all NCF-T samples. A benchmark comparison with representative nitrogen-doped carbon-based electrodes is summarized in Table S2. Although direct comparison is influenced by differences in testing conditions and electrolytes, NCF-900 exhibits competitive electrochemical performance and cycling stability among recently reported nitrogen-doped carbon materials.

Overall, these structural and chemical characteristics are believed to underpin the superior electrochemical performance of NCF-900 and may offer useful design guidance for future high-performance carbon-based supercapacitor electrodes. To demonstrate the practical applicability of the NCF-900 electrodes, symmetric supercapacitor devices were connected in series and evaluated at the device level. As shown in Figure 6a, 6b, the operating voltage increased proportionally with the number of connected cells while maintaining capacitive behavior. Furthermore, three NCF-900 devices connected in series were able to power a commercial red LED for over 3 min after charging (Figure 6c and Movie S1), demonstrating the feasibility of the fabricated electrodes for practical energy-storage applications.

3.3. DFT calculations

DFT calculations were performed to evaluate the proton adsorption behavior on different nitrogen configurations (see Supporting Information for computational details). The adsorption energy (E_{ads}) was used to describe the interaction strength between nitrogen sites and protons. In contrast, the formation energy (E_{form}) reflects the thermodynamic stability of each nitrogen configuration. The calculated formation energies of representative nitrogen configurations (Figure S8) indicate that the considered structures are energetically feasible under the given conditions.

As shown in Figure 5a, the proton adsorption behavior is highly dependent on the nitrogen configuration within the carbon lattice (Figure S9). Among the four nitrogen species examined, pyridinic-N exhibits by far the strongest proton adsorption, followed by oxidized-N, whereas pyrrolic-N and graphitic-N show negligible or unstable proton adsorption [45].

A clear correlation is observed between adsorption strength and the charge state of the adsorbed proton [45]. The stronger adsorption is associated with a more pronounced proton-like character of the adsorbed proton, with its positive charge approaching that of H^+ [41], [45], [46]. After proton adsorption on each nitrogen sites, only the pyridinic-N site directly accepts electron density upon proton adsorption, as evidenced by both partial charge density at the HOMO levels and Bader charge analysis (Figure 5b, 5c). This behavior reflects the strong Lewis basic character of pyridinic-N, which enables localized electron uptake and stabilizes the proton through a direct N–H interaction.

In contrast, for pyrrolic-N and graphitic-N, the transferred electron density is not localized on the nitrogen atom but instead becomes broadly delocalized over the surrounding carbon framework [47]. Such delocalization weakens site-specific interactions with proton and accounts for the absence of stable proton adsorption on these configurations. For oxidized-N, the oxygen partially participates in both proton binding and electron acceptance, resulting in a moderate adsorption strength that is weaker than that of pyridinic-N but stronger than that of pyrrolic and graphitic nitrogen [48]. Also, Proton adsorption at carbon sites neighboring each nitrogen site was evaluated to assess the relative stability of non-nitrogen adsorption pathways. The weaker adsorption at adjacent carbon sites, compared to direct adsorption at pyridinic-N, further support the site-specific preference for proton binding at pyridinic nitrogen (Figure S10).

Further insight into the origin of these adsorption behaviors is provided by the projected density of states (PDOS) analysis (Figure 5d). Pyridinic-N introduces a strongly localized nitrogen-derived electronic state near the Fermi level, which becomes significantly involved upon proton adsorption [49]. This localized state facilitates efficient electron transfer between the adsorbed proton and the carbon framework, thereby stabilizing the protonated species [49]. In contrast, pyrrolic-N and graphitic-N lack such localized states near the Fermi level, while oxidized-N shows partial electronic contributions from oxygen-derived states rather than nitrogen-centered orbitals [37].

Collectively, these results suggest that pyridinic nitrogen is particularly favorable the key requirements for pseudocapacitive redox activity, including strong and stable proton adsorption, localized electron transfer at the nitrogen site, and favorable electronic states near the Fermi level. The DFT results provide theoretical insight consistent with the experimental observation that NCF-900 delivers superior electrochemical performance, owing to its favorable nitrogen configuration together with defect-rich and mesoporous structural characteristics. These theoretical findings provide mechanistic insight into the relative electrochemical activity of different nitrogen species and suggest that pyridinic-N is a major electrochemically active nitrogen configuration contributing to charge storage in nitrogen-doped carbon materials.

4. Conclusions

In this work, a solid-state approach using exfoliated g-C_3N_4 nanosheets as both a nitrogen source and porogen was developed to precisely control the nitrogen configurations in carbon fabrics. By adjusting the pyrolysis temperature, the relative proportions of pyridinic, pyrrolic, and graphitic nitrogen were systematically tuned while preserving the textile-like morphology of the carbon framework. Among the series, NCF-900 exhibited the most favorable combination of electrochemically accessible pyridinic-N sites, well-developed mesoporosity, and abundant structural defects. These synergistic features resulted in enhanced capacitance, rapid charge transport, and excellent cycling stability. Complementary DFT calculations provided additional insight into the charge-storage mechanism. The adsorption energy and charge-density analyses revealed that pyridinic nitrogen sites possess the most favorable proton-binding energy and exhibit pronounced charge redistribution near the Fermi level, indicating their strong tendency to participate in reversible redox reactions. These theoretical results are consistent with the experimental observations and identify pyridinic-N as a major electrochemically active nitrogen configuration contributing to charge-storage behavior. Overall, this study provides experimental and theoretical insight into the relationship between nitrogen configuration, electronic structure, and electrochemical behavior in nitrogen-doped carbon fabrics. The g-C_3N_4-assisted solid-state route presented here offers a scalable, gas-free, and environmentally benign pathway for fabricating flexible carbon electrodes with tunable surface chemistry. The insights gained from both experiment and theory provide a rational basis for the design of next-generation nitrogen-doped carbon materials for high-performance supercapacitors and related energy storage devices.

Conflict of interest

The authors declare no conflict of interest.

Supporting Information

The Supporting Information is available free of charge at

- Photographs of the fabrication process; EDS spectra, elemental mapping, and cross-sectional EDS analysis of NCF-900; thermogravimetric analysis of the g-C_3N_4-coated precursor; schematic illustration of g-C_3N_4 nitrogen bonding motifs and corresponding XPS signatures; high-resolution C 1s XPS spectra; SEM images of NCF-T samples prepared at different pyrolysis temperatures; electrochemical characterization in symmetric two-electrode and three-electrode configurations; DFT models, formation energies, proton adsorption energies, charge-density analysis, and PDOS calculations; Raman ID/IG analysis; comparison of electrochemical performance with representative nitrogen-doped carbon materials; supporting references (PDF).

- Supplementary Movie S1 shows the practical demonstration of the fabricated flexible supercapacitor powering an LED.

Acknowledgments

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References

[1] Simon, P., & Gogotsi, Y. (2008). Materials for electrochemical capacitors. Nature Materials, 7, 845–854.

[2] Zhu, Y., Murali, S., Stoller, M. D., et al. (2011). Carbon-based supercapacitors produced by activation of graphene. Science, 332, 1537–1541.

[3] Zhai, Y., Dou, Y., Zhao, D., Fulvio, P. F., Mayes, R. T., & Dai, S. (2011). Carbon materials for chemical capacitive energy storage. Advanced Materials, 23, 4828–4850.

[4] Shao, Y., El-Kady, M. F., Sun, J., Li, Y., Zhang, Q., Zhu, M., Wang, H., Dunn, B., & Kaner, R. B. (2018). Design and mechanisms of asymmetric supercapacitors. Chemical Reviews, 118, 9233–9280.

[5] Zhang, L., Zhou, R., & Zhao, X. S. (2012). Graphene-based materials as supercapacitor electrodes. Journal of Materials Chemistry, 22, 23468–23484.

[6] Sevilla, M., & Fuertes, A. B. (2012). Sustainable porous carbons with a superior performance for CO_2 capture. Energy Environ. Sci., 5, 5609–5616.

[7] Lin, Z., Goikolea, E., Balducci, A., Naoi, K., Taberna, P. L., Salanne, M., Yushin, G., & Simon, P. (2018). Materials for supercapacitors: When Li-ion battery power is not enough. Materials Today, 21, 419–436.

[8] Li, M., Xu, F., Li, H., & Wang, Y. (2016). Nitrogen-doped porous carbon materials: promising catalysts or catalyst supports for heterogeneous hydrogenation and oxidation. Catalysis Science & Technology, 6, 3670–3693.

[9] Liang, J., Jiao, Y., Jaroniec, M., & Qiao, S. Z. (2012). Sulfur and nitrogen dual-doped mesoporous graphene electrocatalyst for oxygen reduction with synergistically enhanced performance. Angewandte Chemie International Edition, 51, 11496–11500.

[10] Yang, S., Feng, X., Ivanovici, S., & Müllen, K. (2010). Fabrication of graphene-encapsulated oxide nanoparticles: towards high-performance anode materials for lithium storage. Angewandte Chemie International Edition, 49, 8408–8411.

[11] Kim, H., Kim, B. G., & Cho, J. (2014). A low-cost, high-performance, and scalable method toward nitrogen-doped graphene-based cathodes for lithium-oxygen batteries. Energy & Environmental Science, 7, 397–404.

[12] Chen, L., Ji, T., Mu, L., & Zhu, J. (2017). Cotton fabric derived hierarchically porous carbon and nitrogen doping for sustainable capacitor electrode. Carbon, 111, 839–848.

[13] Wang, Q., Yan, J., Wang, Y., Wei, T., Zhang, M., Jing, X., & Fan, Z. (2014). Three-dimensional flower-like and hierarchical porous carbon materials as high-rate performance electrodes for supercapacitors. Carbon, 67, 119–127.

[14] Wu, Z. S., Chen, L., Sun, Y., Muellen, K., Winter, A., Turchanin, A., & Feng, X. (2012). Three-dimensional nitrogen and boron co-doped graphene for high-performance all-solid-state supercapacitors. Advanced Materials (Weinheim), 24.

[15] Li, Z., Yang, Y., Ding, G., Wei, L., Yao, G., Niu, H., Zheng, F., & Chen, Q. (2022). Optimizing the nitrogen configuration in interlayer-expanded carbon materials via sulfur-bridged bonds toward remarkable energy storage performances. Journal of Materials Chemistry A, 10, 10033–10042.

[16] Kresse, G., & Hafner, J. (1993). Ab initio molecular dynamics for liquid metals. Physical Review B, 47, 558.

[17] Perdew, J. P., Burke, K., & Ernzerhof, M. (1996). Generalized gradient approximation made simple. Physical Review Letters, 77, 3865.

[18] Henkelman, G., Arnaldsson, A., & Jónsson, H. (2006). A fast and robust algorithm for Bader decomposition of charge density. Computational Materials Science, 36, 354-360.

[19] Arjona, L., Martín-Lara, M. Á., Montero, Á., Calero, M., Solís, R. R., & Blázquez, G. (2025). Thermogravimetric analysis of textile waste: kinetic parameters and pyrolytic decomposition mechanisms. Thermal Science and Engineering Progress, 103946.

[20] Saeed, M., Munir, M., Intisar, A., & Waseem, A. (2022). Facile synthesis of a novel Ni-WO3@ g-C3N4 nanocomposite for efficient oxidative desulfurization of both model and real fuel. ACS Omega, 7, 15809–15820.

[21] Guo, D., Shibuya, R., Akiba, C., Saji, S., Kondo, T., & Nakamura, J. (2016). Active sites of nitrogen-doped carbon materials for oxygen reduction reaction clarified using model catalysts. Science, 351, 361–365.

[22] Zheng, Y., Jiao, Y., Li, L. H., Xing, T., Chen, Y., Jaroniec, M., & Qiao, S. Z. (2014). Toward design of synergistically active carbon-based catalysts for electrocatalytic hydrogen evolution. ACS Nano, 8, 5290–5296.

[23] Qu, L., Liu, Y., Baek, J. B., & Dai, L. (2010). Nitrogen-doped graphene as efficient metal-free electrocatalyst for oxygen reduction in fuel cells. ACS nano, 4, 1321–1326.

[24] Majdoub, M., Amedlous, A., Anfar, Z., & Jada, A. (2022). Engineering of H-Bonding Interactions in PVA/g-C3N4 Hybrids for Enhanced Structural, Thermal, and Mechanical Properties: Toward Water-Responsive Shape Memory Nanocomposites. Advanced Materials Interfaces, 9, 2200170.

[25] Ambaye, A. D., Zikalala, S. A., Mashiloane, K. C., Nure, J. F., Kebede, M. A., Mokrani, T., & Nxumalo, E. N. (2024). Development of engineered Zn-MOF/g-C3N4 based photoelectrochemical system for real-time sensors and removal of naproxen in wastewater. Talanta Open, 10, 100371.

[26] Wang, X., Li, X., Zhang, L., Yoon, Y., Weber, P. K., Wang, H., Guo, J., & Dai, H. (2009). N-doping of graphene through electrothermal reactions with ammonia. Science, 324, 768–771.

[27] Li, Z. Q., Lu, C. J., Xia, Z. P., Zhou, Y., & Luo, Z. (2007). X-ray diffraction patterns of graphite and turbostratic carbon. Carbon, 45, 1686–1695.

[28] Wang, M., Yang, J., Liu, S., Li, M., Hu, C., & Qiu, J. (2020). Nitrogen-doped hierarchically porous carbon nanosheets derived from polymer/graphene oxide hydrogels for high-performance supercapacitors. Journal of Colloid and Interface Science, 560, 69–76.

[29] Dong, G., Zhang, Y., Pan, Q., & Qiu, J. (2014). A fantastic graphitic carbon nitride (g-C3N4) material: electronic structure, photocatalytic and photoelectronic properties. Journal of Photochemistry and Photobiology C: Photochemistry Reviews, 20, 33–50.

[30] Lin, T., Chen, I. W., Liu, F., Yang, C., Bi, H., Xu, F., & Huang, F. (2015). Nitrogen-doped mesoporous carbon of extraordinary capacitance for electrochemical energy storage. Science, 350, 1508–1513.

[31] Sevilla, M., & Fuertes, A. B. (2009). The production of carbon materials by hydrothermal carbonization of cellulose. Carbon, 47, 2281–2289.

[32] Toth, P. (2021). Nanostructure quantification of turbostratic carbon by HRTEM image analysis: State of the art, biases, sensitivity and best practices. Carbon, 178, 688–707.

[33] Li, J., Gao, Y., Han, K., Qi, J., Li, M., & Teng, Z. (2019). High performance hierarchical porous carbon derived from distinctive plant tissue for supercapacitor. Scientific Reports, 9, 17270.

[34] Laschuk, N. O., Easton, E. B., & Zenkina, O. V. (2021). Reducing the resistance for the use of electrochemical impedance spectroscopy analysis in materials chemistry. RSC Advances, 11, 27925–27936.

[35] Ilnicka, A., Skorupska, M., Szkoda, M., Zarach, Z., & Lukaszewicz, J. P. (2023). N-doped carbon materials as electrodes for highly stable supercapacitors. Materials Research Letters, 11, 213–221.

[36] Roy, N., Yasmin, S., Mohiuddin, A. K., & Jeon, S. (2023). Pyridinic and Pyrrolic-N induced carbon nanotubes support bimetallic oxide nanoparticles as high-performance electrocatalysts for oxygen reduction reaction. Applied Surface Science Advances, 18, 100517.

[37] Li, J., Wang, R., Han, L., Wang, T., Asakura, Y., Wang, C., Wang, G., Xu, X., & Yamauchi, Y. (2025). Unveiling the neglected role of oxygen doping in nitrogen-doped carbon for enhanced capacitive deionization performance. Nature Communications, 16, 1996.

[38] Qiao, M., Tang, C., He, G., Qiu, K., Binions, R., Parkin, I. P., Zhang, Q., Guo, Z., & Titirici, M. M. (2016). Graphene/nitrogen-doped porous carbon sandwiches for the metal-free oxygen reduction reaction: conductivity versus active sites. Journal of Materials Chemistry A, 4, 12658–12666.

[39] Ilnicka, A., Skorupska, M., Szkoda, M., Zarach, Z., Kamedulski, P., Zielinski, W., & Lukaszewicz, J. P. (2021). Combined effect of nitrogen-doped functional groups and porosity of porous carbons on electrochemical performance of supercapacitors. Scientific Reports, 11, 18387.

[40] Mirzaeian, M., Abbas, Q., Hunt, M. R., & Hall, P. (2020). Pseudocapacitive effect of carbons doped with different functional groups as electrode materials for electrochemical capacitors. Energies, 13, 5577.

[41] Faisal, S. N., Haque, E., Noorbehesht, N., Zhang, W., Harris, A. T., Church, T. L., & Minett, A. I. (2017). Pyridinic and graphitic nitrogen-rich graphene for high-performance supercapacitors and metal-free bifunctional electrocatalysts for ORR and OER. RSC Advances, 7, 17950–17958.

[42] Tian, K., Wang, J., Cao, L., Yang, W., Guo, W., Liu, S., Li, W., Wang, F., Li, X., Xu, Z., Wang, Z., Wang, H., & Hou, Y. (2020). Single-site pyrrolic-nitrogen-doped sp 2-

hybridized carbon materials and their pseudocapacitance. Nature Communications, 11, 3884.
[43] Yutomo, E. B., Noor, F. A., & Winata, T. (2021). Effect of the number of nitrogen dopants on the electronic and magnetic properties of graphitic and pyridinic N-doped graphene–a density-functional study. RSC Advances, 11, 18371–18380.
[44] Luo, X. Y., Chen, Y., & Mo, Y. (2021). A review of charge storage in porous carbon-based supercapacitors. New Carbon Materials, 36, 49–68.
[45] Guo, D., Shibuya, R., Akiba, C., Saji, S., Kondo, T., & Nakamura, J. (2016). Active sites of nitrogen-doped carbon materials for oxygen reduction reaction clarified using model catalysts. Science, 351, 361–365.
[46] Bulusheva, L. G., Fedorovskaya, E. O., Kurenya, A. G., & Okotrub, A. V. (2013). Supercapacitor performance of nitrogen-doped carbon nanotube arrays. Physica Status Solidi (B), 250, 2586–2591.
[47] Qiao, M., Wang, Y., Wang, Q., Hu, G., Mamat, X., Zhang, S., & Wang, S. (2016). Graphene/nitrogen-doped porous carbon sandwiches for the metal-free oxygen reduction reaction. Journal of Materials Chemistry A, 4, 12658–12666.
[48] Wu, J., Ma, L., Yadav, R. M., Yang, Y., Zhang, X., Vajtai, R., Lou, J., & Ajayan, P. M. (2015). Nitrogen-doped graphene with pyridinic dominance as a highly active and stable electrocatalyst for oxygen reduction. ACS Applied Materials & Interfaces, 7, 14763–14769.
[49] Jiao, Y., Zheng, Y., Jaroniec, M., & Qiao, S. Z. (2014). Origin of the electrocatalytic oxygen reduction activity of graphene-based catalysts: a roadmap to achieve the best performance. Journal of the American Chemical Society, 136, 4394–4403.

Figures

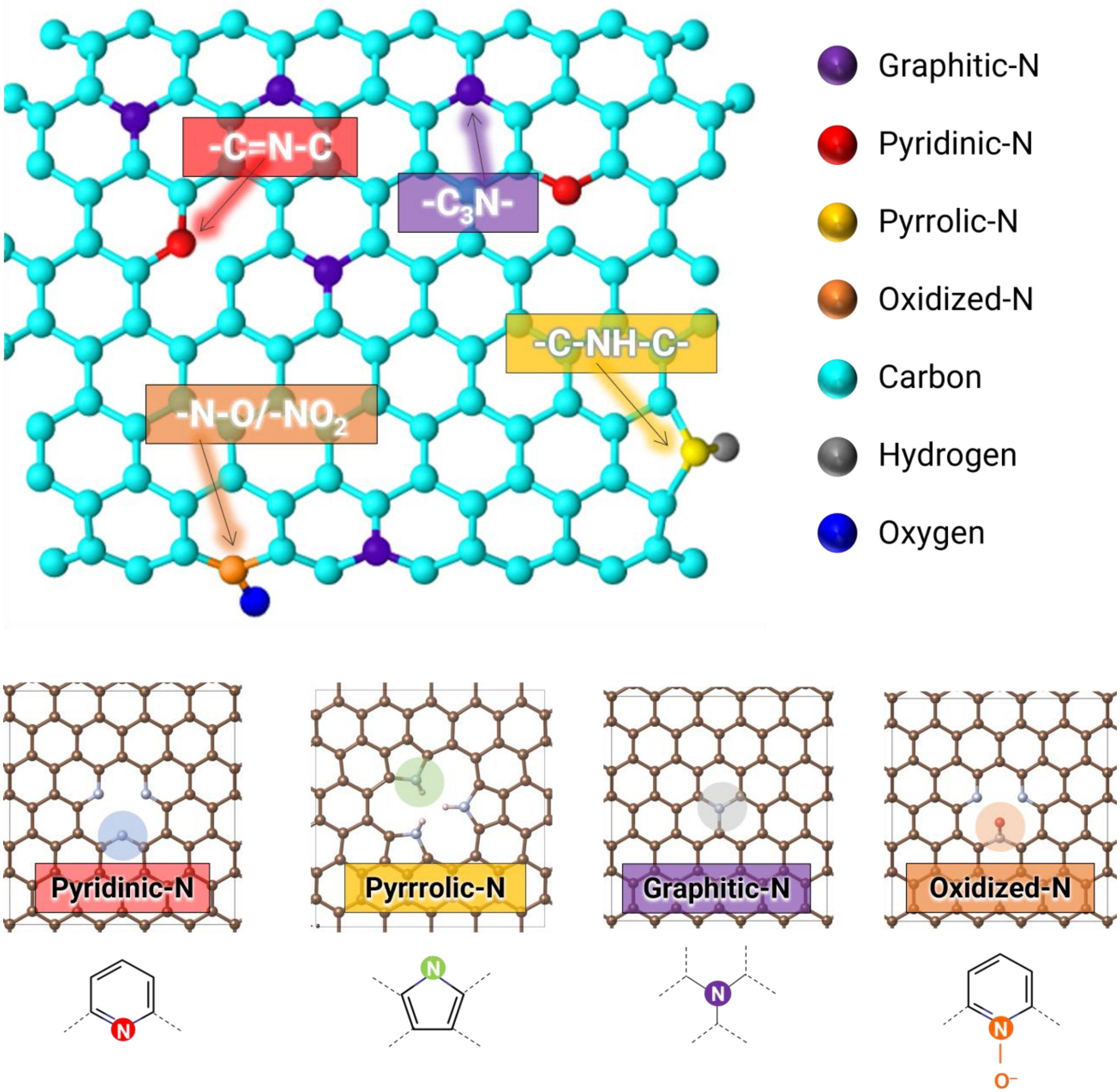

Figure 1. Schematic representation of the four major nitrogen configurations incorporated into the carbon lattice. (1) pyridinic-N (–C=N–C) located at the edge of six-membered rings contributing one p-electron to the π-system, (2) pyrrolic-N (–C–NH–C–) within five-membered rings bonded to two carbon atoms and one hydrogen atom, (3) graphitic-N (–C_3N–) substituting a carbon atom within the graphene basal plane, and (4) oxidized-N (–N–O / –NO_2) bonded to oxygen-containing groups at defect sites. This schematic highlights the distinct bonding environments and potential electronic effects of each nitrogen type on the carbon framework.

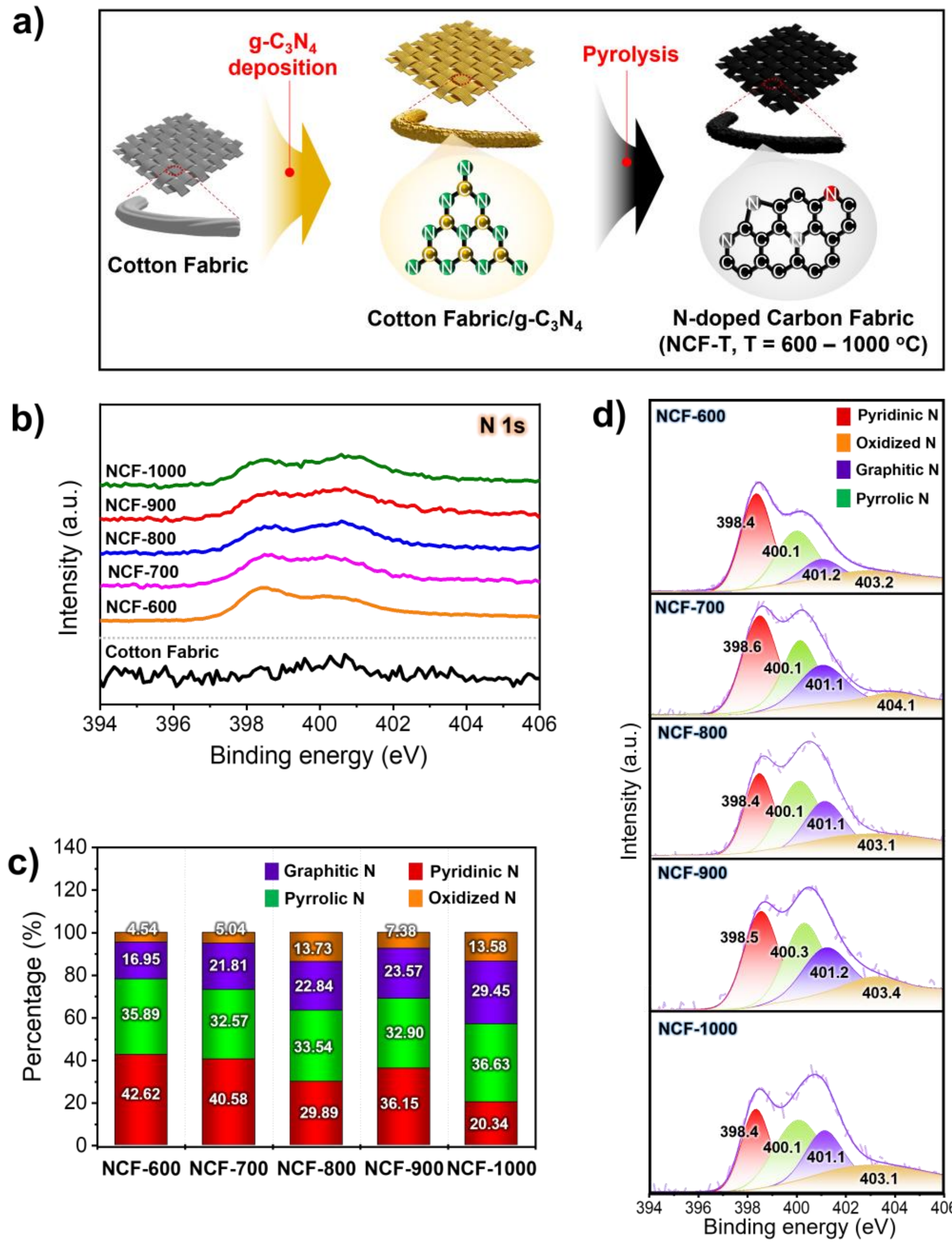

Figure 2. Schematic illustration and chemical characterization of nitrogen-doped carbon fabrics (NCF-T, T = 600–1000 °C) synthesized through a g-C_3N_4-assisted solid-state route. **a)** Schematic representation of the fabrication process, showing exfoliated g-C_3N_4 nanosheets deposited on cotton fabric and subsequently carbonized to form porous nitrogen-doped carbon fibers; **b)** XPS survey spectra of NCF-T, showing the presence of N 1s signals in all nitrogen-doped carbon fabrics; **c)** Relative proportions of nitrogen configurations, including pyridinic, pyrrolic, graphitic, and oxidized nitrogen, extracted from deconvolution of N 1s high-resolution spectra; **d)** High-resolution N 1s XPS spectra of NCF-T after peak deconvolution, illustrating the temperature-dependent evolution of pyridinic, pyrrolic, graphitic, and oxidized nitrogen species.

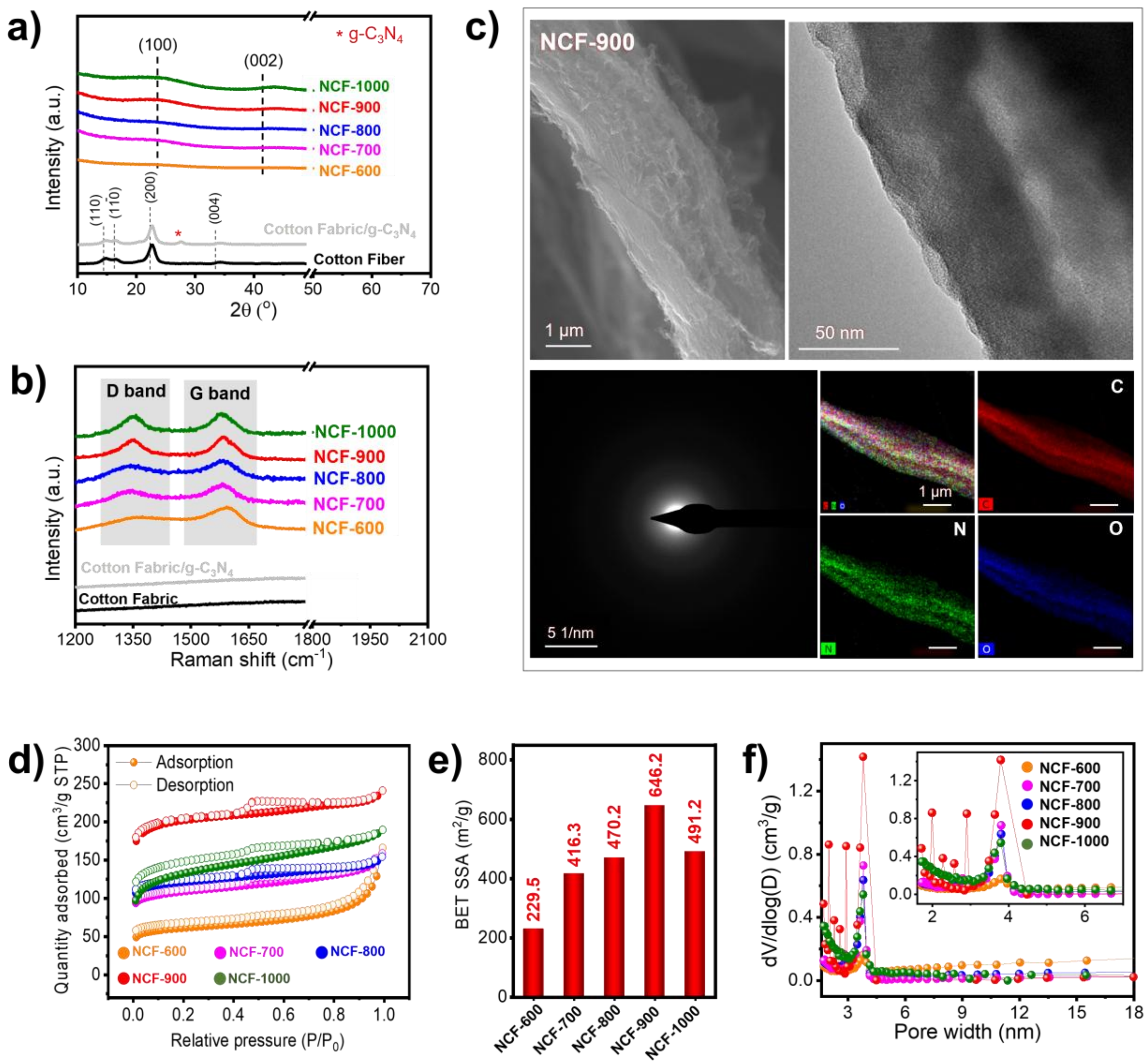

Figure 3. Structural and morphological characterization of cotton fabric, cotton/g-C_3N_4 precursor, and NCF-T samples (T = 600–1000 °C). **a)** XRD patterns showing the gradual evolution from amorphous carbon to partially graphitized domains with increasing pyrolysis temperature; **b)** Raman spectra displaying D and G bands, where the highest I_D/I_G ratio in NCF-900 indicates the highest defect density associated with nitrogen incorporation; **c)** SEM, TEM, and SAED images of NCF-900 revealing a porous and partially ordered carbon framework. TEM–EDS (energy dispersive x-ray spectroscopy) elemental mapping confirms the presence of C, N, and O distribution within the carbon framework; **d)** Nitrogen adsorption–desorption isotherms of NCF-T samples showing type IV behavior with distinct hysteresis loops characteristic of mesoporous structures; **e)** BET-specific surface areas of NCF-T samples, revealing the maximum surface area of ~646 $m^2\ g^{-1}$ for NCF-900 compared to other temperatures, which supports its superior electrochemical performance. **f)** BJH (Barrett, Joyner, Helenda) pore-size distributions confirming mesopores centered at ~3.9 nm for NCF-900, favorable for electrolyte ion transport. Cumulative pore-volume plots indicate the highest total pore volume for NCF-900, consistent with its enhanced mesoporosity and ion accessibility.

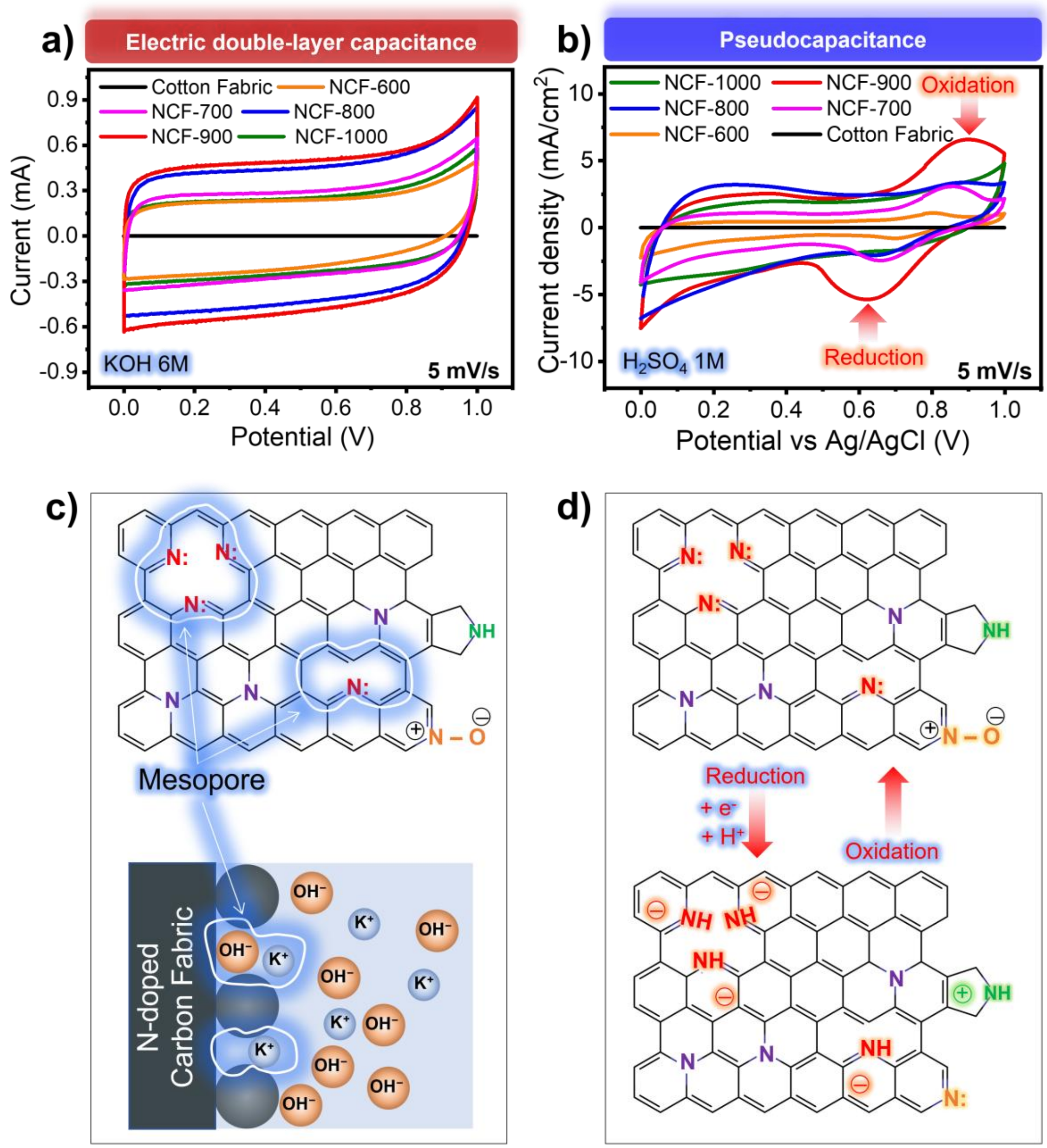

Figure 4. Electrochemical characterization and proposed charge-storage mechanism of nitrogen-doped carbon fabrics. CV curves of cotton fabric and NCF-T measured in **a)** symmetric two-electrode devices using 6 M KOH electrolyte at 5 mV s^{-1} and **b)** a three-electrode configuration using 1 M H_2SO_4 electrolyte at 5 mV s^{-1}; **c)** Schematic illustration of the proposed relationship between nitrogen configurations, defect generation, mesoporosity development, and ion transport; **d)** Proposed charge-storage mechanism associated with nitrogen functionalities. Pyridinic and pyrrolic nitrogen species may contribute to pseudocapacitive behavior through surface redox reactions and proton-coupled electron-transfer

processes, while graphitic nitrogen primarily enhances electronic conductivity.

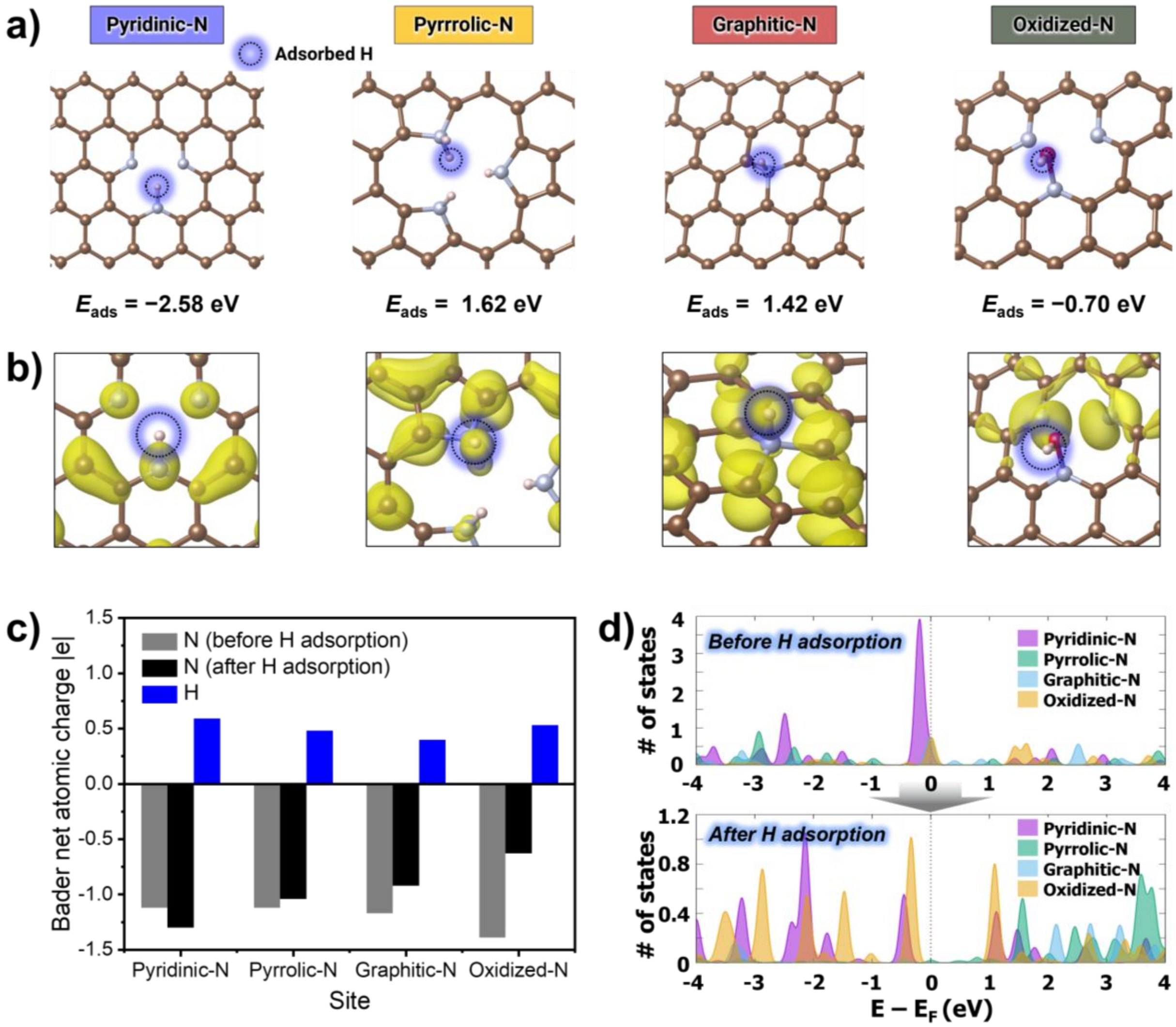

Figure 5. Density functional theory (DFT) analysis of proton adsorption on different nitrogen configurations in carbon. **a)** Optimized adsorption geometries and calculated proton adsorption energies (E_{ads}) for pyridinic-N, pyrrolic-N, graphitic-N, and oxidized-N sites. **b)** Partial charge density at the HOMO level of after proton adsorption on each nitrogen sites. **c)** Bader charge analysis comparing the charge states of nitrogen atoms before and after proton adsorption, indicating pronounced electron uptake at pyridinic-N sites. **d)** Projected density of states (PDOS) of nitrogen before and after proton adsorption, showing the emergence of localized electronic states near the Fermi level for pyridinic-N.

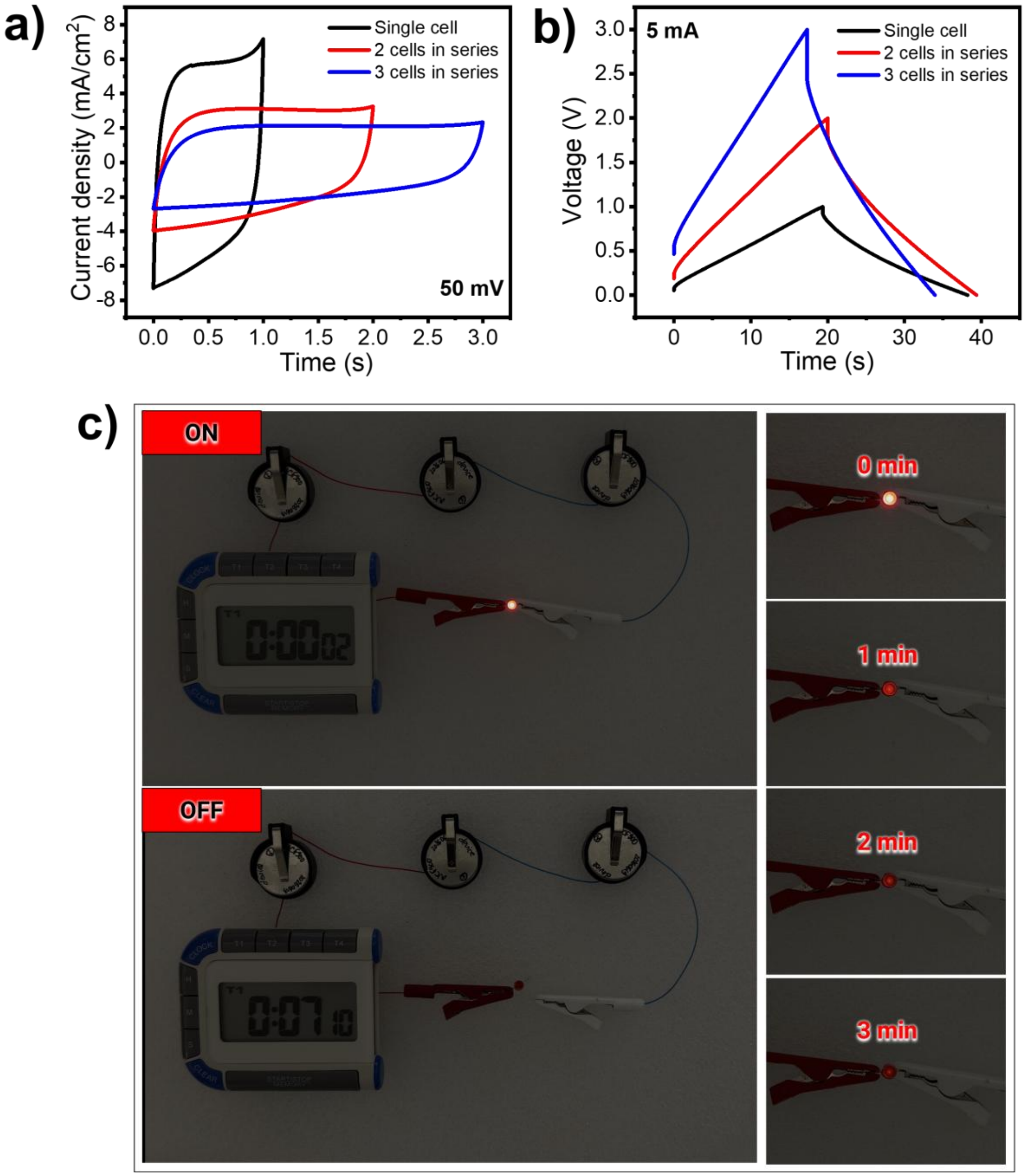

Figure 6. Device-level performance and practical demonstration of NCF-900 symmetric supercapacitors connected in series. **a)** CV curves and **(b)** GCD profiles of one, two, and three cells connected in series; **c)** Photographs demonstrating the illumination of a red LED powered by three series-connected NCF-900 devices and the corresponding discharge process.

Tables

Table 1. Summary of representative nitrogen configurations identified in nitrogen-doped carbon materials, including their bonding environment, characteristic N 1s binding-energy ranges, and schematic molecular fragments. The four main configurations—pyridinic-N, pyrrolic-N, graphitic-N, and oxidized-N—exhibit distinct electronic and structural roles within the carbon lattice.

Type of N	Structure / Bonding	Binding energy (N 1s, eV)	Schematic fragment
Pyridinic-N	N atom bonded to two C atoms at the edge of a six-membered ring; contributes one p-electron to aromatic π-system	398.3 – 398.8	–C=N–C– (at graphene edge)
Pyrrolic-N	N atom in a five-membered ring, bonded to two C atoms and one H atom	399.8 – 400.5	–C–NH–C– (pyrrole-like)
Graphitic-N	N atom substituting C within the graphene plane, bonded to three C atoms in sp^2 configuration	400.8 – 401.2	$–C_3N–$ (in-plane)
Oxidized-N	N atom bonded to carbon structures that have been chemically altered, often by losing electrons to oxygen	402.5 – 405.0	$–NO^-$ (at graphene edge)

Table 2. Relative fractions (%) of nitrogen configurations (pyridinic-N, pyrrolic-N, graphitic-N, and oxidized-N) obtained from high-resolution N 1s XPS spectra of NCF-T samples.

Sample	Pyridinic-N (%)	Pyrrolic-N (%)	Graphitic-N (%)	Oxidized-N (%)
Cotton Fabric	-	-	-	-
NCF-600	42.62	35.57	17.27	4.54
NCF-700	40.58	32.57	21.81	5.04
NCF-800	29.89	33.54	22.84	13.73
NCF-900	36.15	32.9	23.57	7.38
NCF-1000	20.34	36.63	29.45	13.58

Table 3. Calculated atomic contents (at.%) of different nitrogen configurations obtained by combining the total nitrogen content from XPS survey spectra with the relative fractions derived from high-resolution N 1s spectra.

Sample	Total N content (at.%)	Calculated Pyridinic-N content (at.%)	Calculated Pyrrolic-N content (at.%)	Calculated Graphitic-N content (at.%)	Calculated Oxidized-N content (at.%)
NCF-600	15.34	6.54	5.46	2.65	0.70
NCF-700	9.05	3.67	2.95	1.97	0.46
NCF-800	8.32	2.49	2.79	1.90	1.14
NCF-900	8.37	3.03	2.75	1.97	0.62
NCF-1000	2.01	0.41	0.74	0.59	0.27

SUPPORTING INFORMATION

Role of Pyridinic Nitrogen in Redox Activity of Nitrogen-Doped Carbon Fabrics

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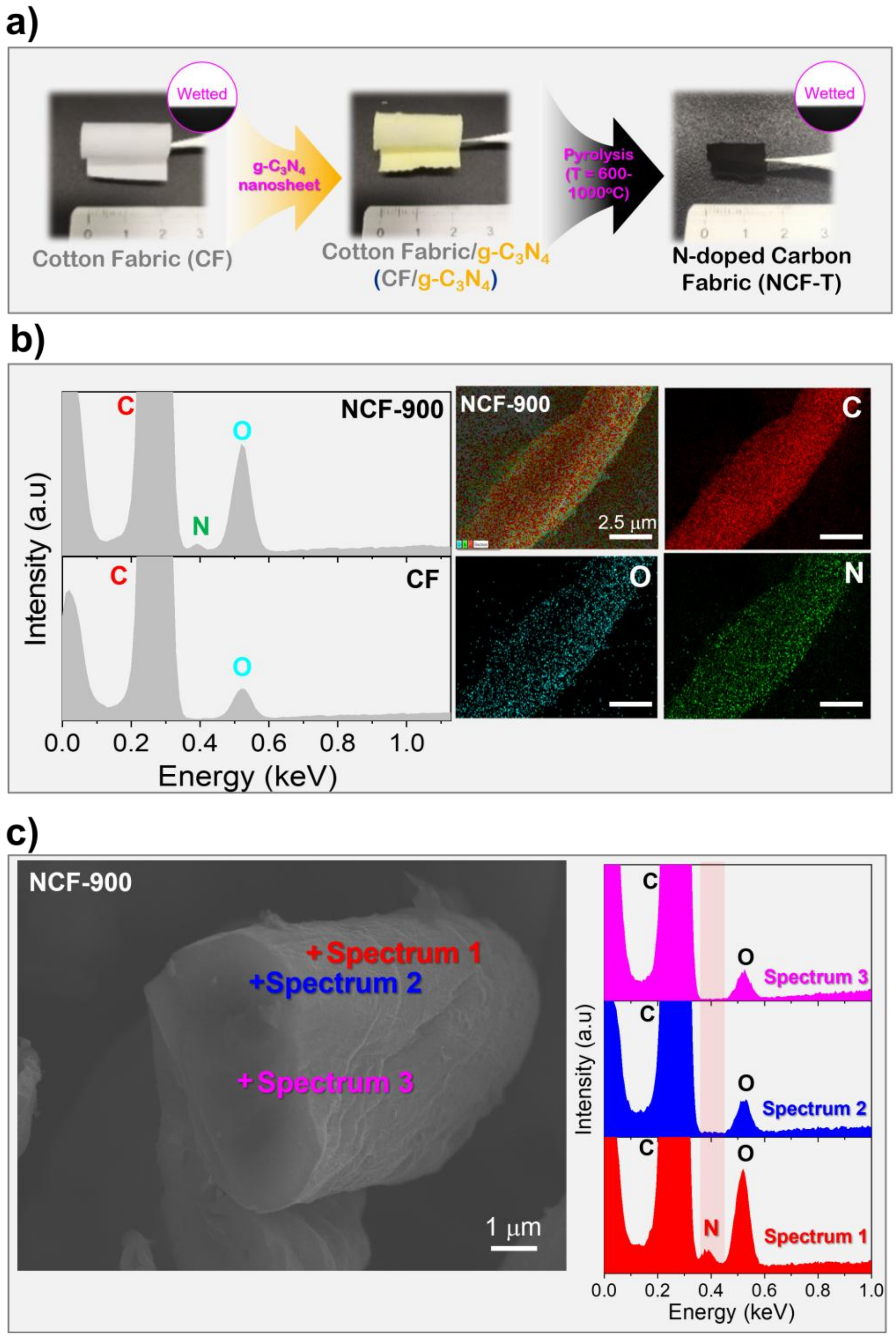

Figure S1. a) Photographs illustrating the fabrication process from pristine cotton fabric (CF) to g-C_3N_4-coated cotton fabric (CF/g-C_3N_4) and nitrogen-doped carbon fabric (NCF-T); **b)** EDS spectra and elemental mapping images of NCF-900; **c)** Cross-sectional SEM image and corresponding EDS point spectra acquired at different positions across the NCF-900 fiber.

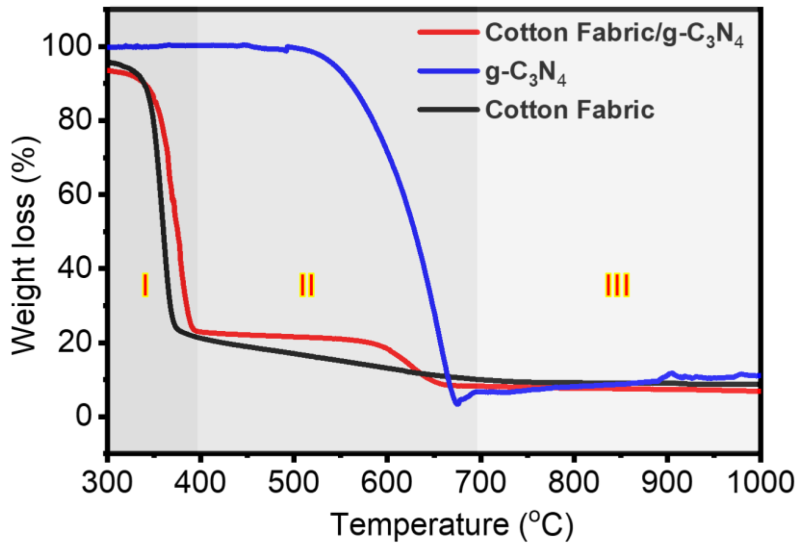

Figure S2. Thermogravimetric analysis (TGA) of g-C_3N_4-coated cotton precursor. TGA of the g-C_3N_4-coated cotton precursor shows two-step thermal degradation corresponding to cellulose decomposition (~350 °C) and g-C_3N_4 breakdown (~600 °C). These transitions confirm the simultaneous evolution of carbon structure and nitrogen doping during pyrolysis.

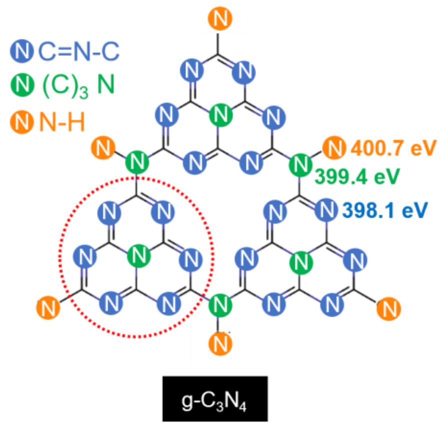

Figure S3. Schematic illustration of the chemical structure of g-C_3N_4 and its characteristic nitrogen bonding motifs, including C=N–C, $(C)_3$–N, and N–H species, together with their corresponding N 1s XPS binding energies at approximately 398.1, 399.4, and 400.7 eV, respectively. These features serve as diagnostic fingerprints for identifying residual g-C_3N_4 species in low-temperature carbonized samples.

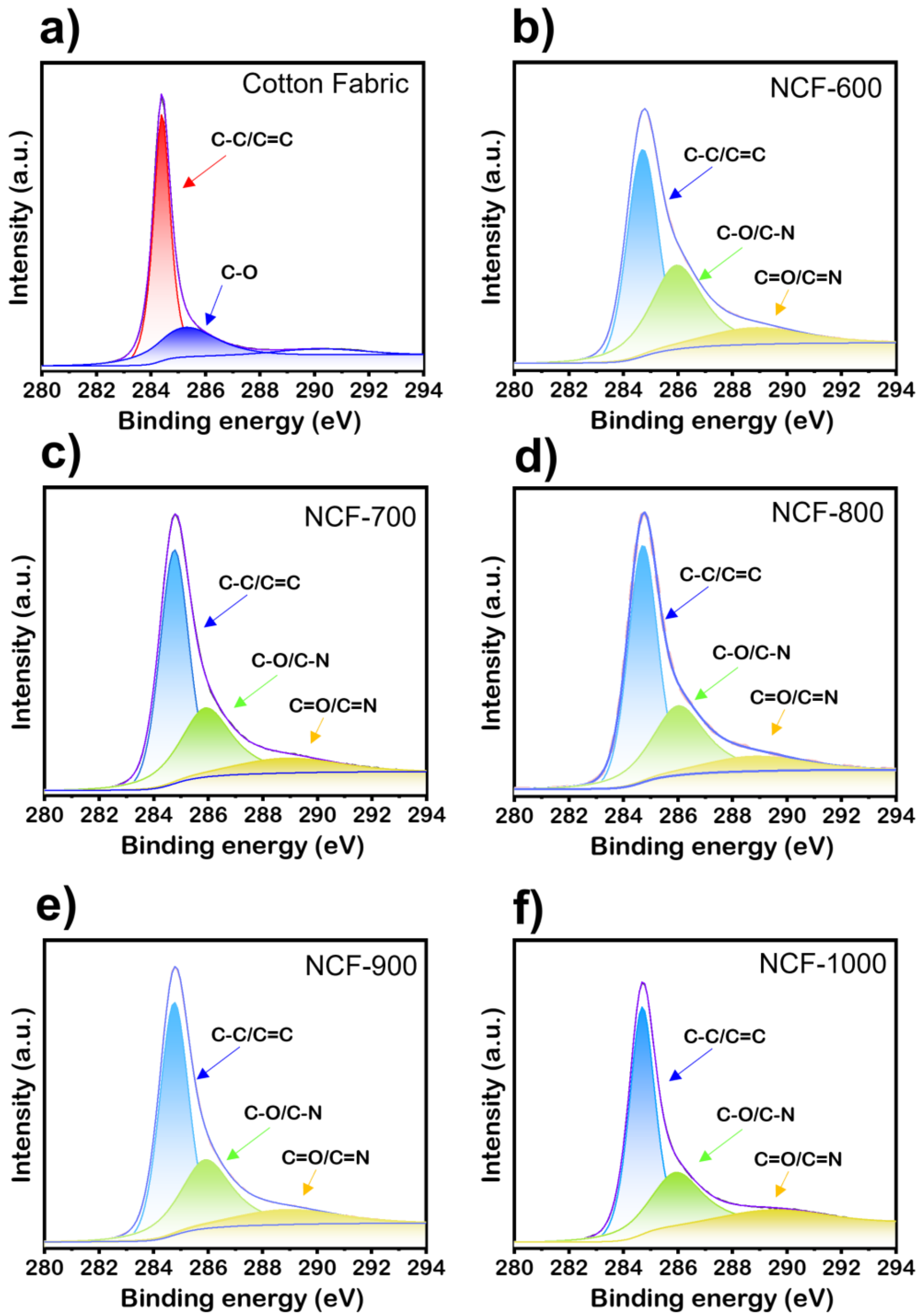

Figure S4. High-resolution C 1s XPS spectra of **a)** Pristine cotton fabric and **b–f)** NCF-T samples carbonized at different temperatures. All spectra can be deconvoluted into three main components corresponding to sp²-hybridized carbon (C–C/C=C, ~284.6 eV), C–O/C–N (~286.1 eV), and C=O/C–O–N (~288.5 eV). The evolution of these peaks with increasing temperature confirms the formation of nitrogen-containing carbon species and the progressive graphitization of the carbon framework.

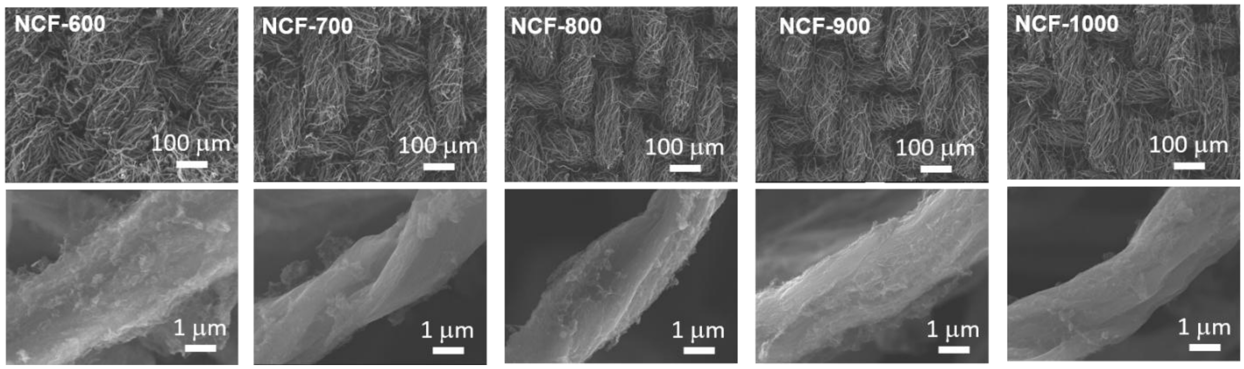

Figure S5. **Morphological evolution of NCF-T samples at different carbonization temperatures.** SEM images show NCF-T samples synthesized at 600–1000 °C. NCF-900 retains a well-defined fibrous morphology with a rough, porous surface, whereas NCF-600 appears under-carbonized, and NCF-1000 exhibits partial pore collapse due to excessive graphitization.

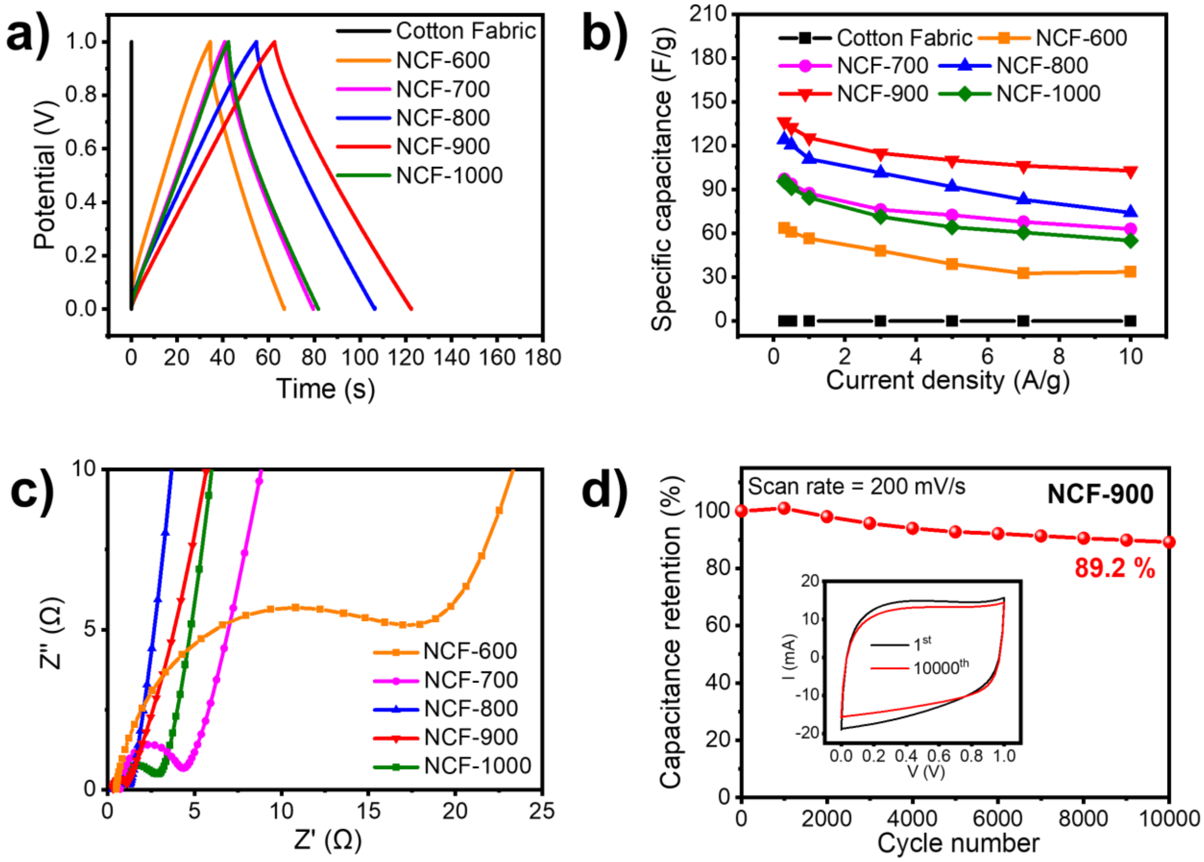

Figure S6. Electrochemical performance in symmetric two-electrode configurationof cotton fabric and NCF-T materials. a) Galvanostatic charge–discharge (GCD) profiles of symmetric two-electrode devices using 6 M KOH electrolyte at 10 mV/s; **b)** Areal capacitance of NCF-T calculated from CV curves at different scan rates in the two-electrode system; **c)** Nyquist plot of symmetric two-electrode devices of NCF-900, showing electrochemical impedance behavior across the frequency range; **d)** Long-term cycling performance of the NCF-900 symmetric device tested in 6 M KOH over 10,000 charge–discharge cycles.

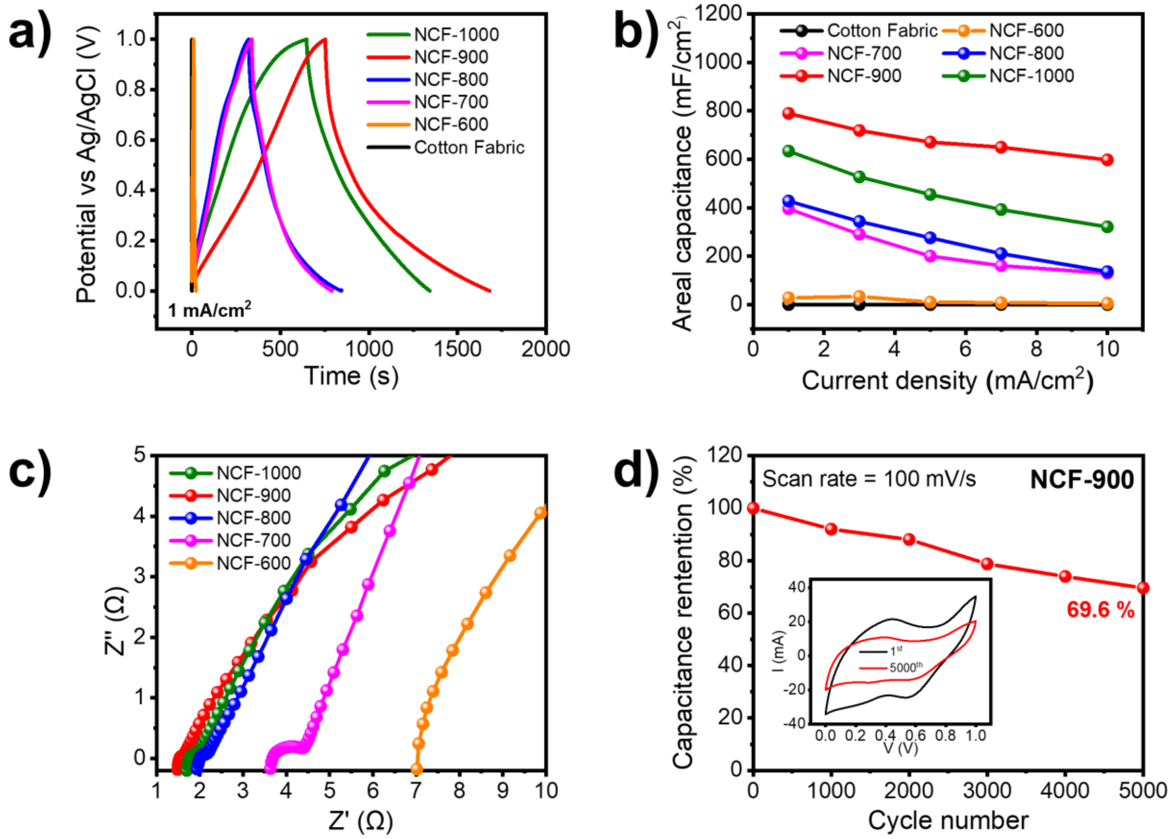

Figure S7. Electrochemical performance in three-electrode configuration of cotton fabric and NCF-T materials. (a) Galvanostatic charge–discharge (GCD) profiles of the samples measured in a three-electrode configuration using 1 M H_2SO_4 electrolyte at 10 mV s^{-1}; **(b)** Areal capacitances of NCF-T calculated from CV curves at different current densities in the three-electrode system; **(c)** Nyquist plots of NCF-T samples showing electrochemical impedance behavior across the frequency range; **(d)** Long-term cycling performance of the NCF-900 symmetric device tested in 1 M H_2SO_4 over 5,000 charge–discharge cycles (inset: representative CV curves at various scan rates).

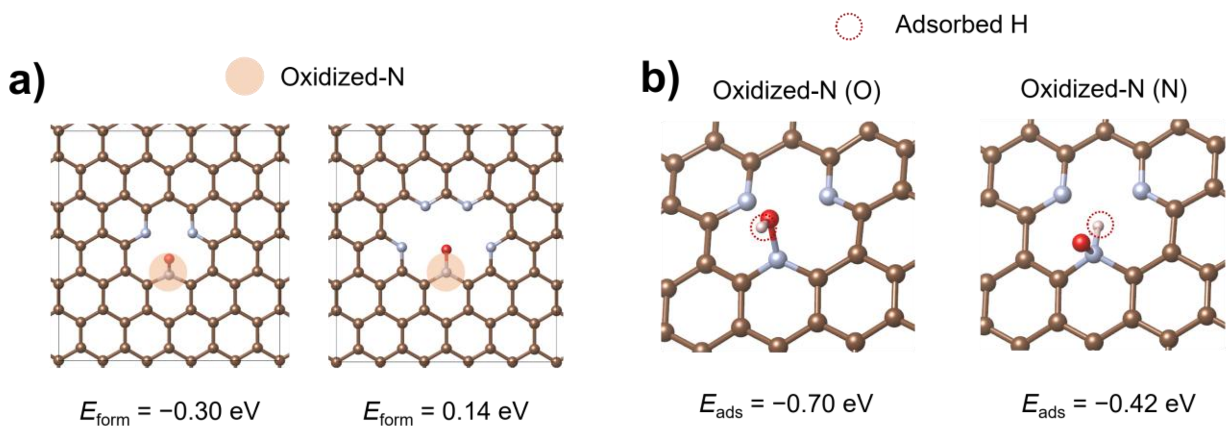

Figure S8. Proton adsorption behavior on oxidized-N sites examined by DFT calculations. **a)** Oxidized-N structures for different carbon defect size and corresponding oxidized-N formation energy (E_{form}, under O-rich conditions). **b)** Calculated proton adsorption energies at the oxygen and nitrogen sites of the most stable oxidized-N structure. In Oxidized-N (O) and Oxidized-N (N), the parentheses indicate the proton adsorption site.

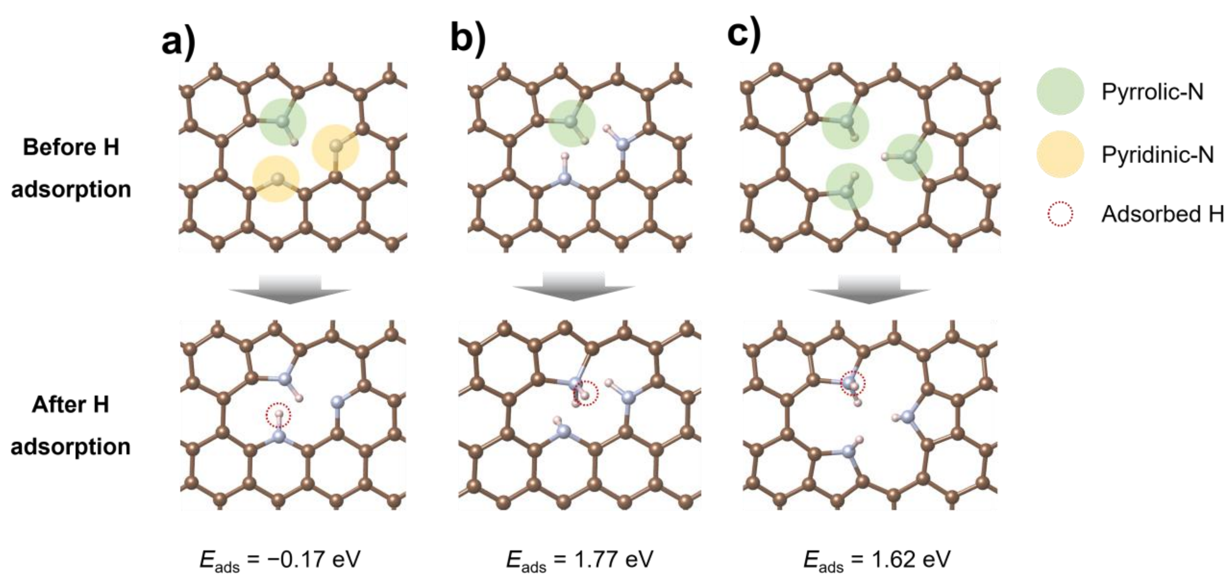

Figure S9. Structural variations of the pyrrolic-N site considered for proton adsorption in the DFT calculations. The pyrrolic-N site in the presence of **a)** pyridinic-N, **b)** hydrogenated pyridine-N, and **c)** pyrrolic-N was examined. After structural relaxation, proton does not form a stable N–H bond at the pyrrolic-N site, indicating weak proton stabilization at pyrrolic-N site under the investigated conditions.

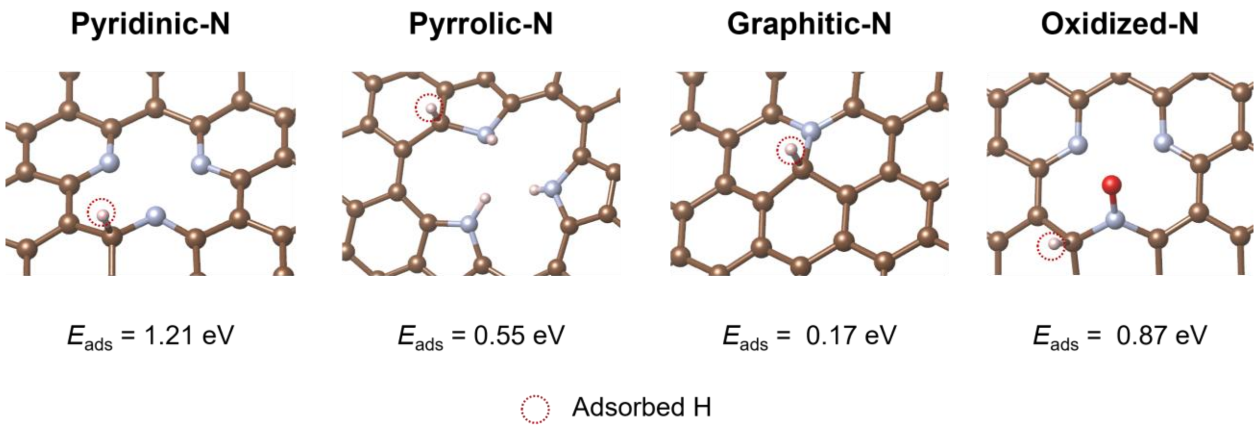

Figure S10. Calculated proton adsorption energies at adjacent carbon sites near different nitrogen configurations.

4. Supporting tables

Table S1. Summary of D- and G-band intensity ratios (I_D/I_G) obtained from the Raman spectra of NCF-T samples prepared at different pyrolysis temperatures. The progressive increase in I_D/I_G up to 900 °C indicates a rise in defect density associated with pyridinic and pyrrolic nitrogen incorporation, while the subsequent decrease at 1000 °C suggests partial defect healing due to graphitization.

Sample	I_D/I_G	Interpretation
NCF-600	0.44	Low I_D/I_G arising from incomplete carbonization and the absence of a well-developed sp^2 carbon network, rather than from a low defect density. The weak D band reflects limited edge sites due to the polymeric nature of residual g-C_3N_4.
NCF-700	0.69	Onset of carbon reconstruction accompanied by initial nitrogen-induced disorder, leading to the emergence of defect-related D-band features.
NCF-800	0.79	Significant increase in structural disorder associated with the formation of abundant edge sites and active nitrogen configurations during carbon framework evolution.
NCF-900	0.87	Highest defect density, indicating an optimal balance between graphitic domains and nitrogen-induced disorder, favorable for electrochemical activity.
NCF-1000	0.76	Partial defect healing due to high-temperature graphitization, resulting in reduced disorder and fewer electrochemically active sites.

Table S2. Comparison of the electrochemical performance of g-C_3N_4-derived NCF-900 with recently reported nitrogen-doped carbon-based electrodes for supercapacitor applications.

Material	Nitrogen Source / Strategy	Electrolyte	Capacitance	Cycling	Ref.
N-doped carbon fabrics	Melamine, cotton fabric / Pyrolysis	6 M KOH	180 F g^{-1} @ 0.5 A g^{-1}	95% (5,000 cycles)	[S1]
N-doped porous carbon	Juncus, $ZnCl_2$ / Pyrolysis	6 M KOH	290.5 F g^{-1} @ 0.5 A g^{-1}	94.5% (10,000 cycles)	[S2]
N-doped carbon	PAN, melamine / Pyrolysis	6 M KOH	135.3 F/g @ 0.5 A g^{-1}	96.6% (5,000 cycles)	[S3]
N-doped carbon	Citric acid, urea / Wet ball milling	1 M Na_2SO_4	79.25 F g^{-1} @ 1 A g^{-1}	96.1% (2,500 cycles)	[S4]
N-doped porous carbon	$ZnCl_2$, NH_4Cl, Ginger straw / Pyrolysis	EMIM-BF_4	122 F g^{-1} @ 0.5 A g^{-1}	87% (10,000 cycles)	[S5]
N-doped porous carbon	Walnut shells, melamine / KOH activation	6 M KOH	329.2 F g^{-1} @ 1 A g^{-1}	90.12 % (5000 cycles)	[S6]
N-doped carbon fabrics	**g-C_3N_4, cotton fabric / Pyrolysis**	**1 M H_2SO_4**	**789.4 mF cm^{-2} @ 1 mA cm^{-2}**	**69.6 % (5000 cycles)**	**This work**
N-doped carbon fabrics	**g-C_3N_4, cotton fabric / Pyrolysis**	**6 M KOH**	**139.6 F g^{-1} @ 0.5 A g^{-1}**	**89% (10,000 cycles)**	**This work**

5. Supporting references

[S1] Chen, L., Ji, T., Mu, L., & Zhu, J. (2017). Cotton fabric derived hierarchically porous carbon and nitrogen doping for sustainable capacitor electrode. Carbon, 111, 839-848.

[S2] He, G., Yan, G., Song, Y., & Wang, L. (2020). Biomass Juncus derived nitrogen-doped porous carbon materials for Supercapacitor and oxygen reduction reaction. Frontiers in Chemistry, 8, 1.

[S3] Fu, X., Li, R., Yan, S., Yuan, W., Zhang, Y., Sun, W., & Wang, X. (2023). Porous carbons prepared from polyacrylonitrile doped with graphitic carbon nitride or melamine for supercapacitor applications. ChemistrySelect, 8, e202301801.

[S4] Devi, M., Shikhar, J., & Sharma, S. (2025). Effect of nitrogen content on performance of supercapacitors composed of nitrogen–carbon materials. Journal of Materials Chemistry A, 13, 42343-42354.

[S5] Zheng, Y., Wang, H., Sun, S., Lu, G., Liu, H., Huang, M., ... & Li, H. (2020). Sustainable nitrogen-doped carbon electrodes for use in high-performance supercapacitors and Li-ion capacitors. Sustainable Energy & Fuels, 4, 1789-1800.

[S6] Niu, C., Ma, X., Zhou, J., Shi, Q., Qu, H., Feng, X., ... & Zhang, R. (2025). Controlled synthesis of pyridinic-enriched nitrogen-doped and high specific surface area porous carbon derived from walnut shell for supercapacitors. Chemical Engineering Science, 122499.